

On one of the applications of Electrodynamics and Crystals

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On light, aka Polarisation:

A mind map on physics of light-matter interaction

I know that title may sound indifferent, and the following description might also appear cumbersome. Therefore, to make things understandable (hopefully), I will follow a tree-style description. All odd numbers are questions and even numbers are respective answers. Our approach is to first construct an intuitive understanding of light in free space (its wave nature, polarisation, and associated physical parameters). Then, we conceptually try to establish the interconnections between these parameters (such as refractive index n , permittivity ϵ , permeability μ , wavevector $\vec{k}$, phase and group velocities), forming a basic idea for light propagation in a vacuum. Next, we extend this framework to light-matter interaction, examining how and why these parameters are modified in different classes of media. In isotropic materials, these quantities remain scalar, whereas in anisotropic materials they become tensorial, because the quantities depend on more parameters than initially thought (eg, polarisation-dependent propagation). Finally, we investigate how varying certain parameters of the incident light, such as frequency, polarisation, or intensity, which can trigger different material responses, ranging from the linear regime to various non-linear optical effects. Our method is to keep asking questions in sequence until we bind a few possible concepts together, hopefully. It may still contain some conceptual errors or typos, as the presentation is based on non-standard ways, and it may appear like a transcript of a talk. Due credits have been given for all images and literary sources used.

0.1 Light, when free and some basics

1. *How is light in free space?*
- 2.

$$\vec{E}(\vec{r}, t) = E_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{E}(\phi)$$

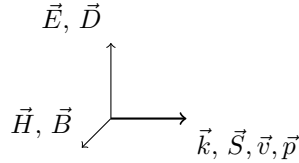
$$\vec{H}(\vec{r}, t) = H_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{H}(\phi)$$

Maxwell has shown (as the one of the solutions of his wave equation) that mutually orthogonal, self-sustaining oscillations of electric field ($\vec{E}$) and magnetic field ($\vec{H}$), propagating in the direction ($\vec{k}$) orthogonal to both oscillating fields [$\vec{k} \times \vec{E} = (\dots)\vec{H}$; $\vec{k} \times \vec{H} = -(\dots)\vec{E}$], is a transverse wave ($\vec{k} \cdot \vec{E} = 0$; $\vec{k} \cdot \vec{H} = 0$), aka electromagnetic wave exists.

3. *Are these only sets of vectors ($\vec{E}$, $\vec{H}$ and $\vec{k}$) will ever be associated the light? And how all look together?*

4. No. First of all, a better term in the place of ‘set of vectors’ is ‘physical quantities’ associated with the light wave. We have electric displacement vector $\vec{D} = (\dots)\vec{E}$, magnetic displacement vector $\vec{B} = (\dots)\vec{H}$, Poynting vector $\vec{S} = (\dots)\vec{E} \times \vec{H}$, momentum vector $\vec{p} = (\dots)\vec{k}$. $\vec{E}$ expresses the direction of electric oscillation, and is also linked to amplitude E_0 , and always in the same as the direction of electric polarisation ($\vec{P} \equiv \hat{E}$), whereas $\vec{P}$ specifically describes electric field oscillation direction and phase ϕ , [similar description exists for $\vec{H}$], $\vec{D}$ expresses the direction of oscillation when $\vec{E}$ is induced in the medium, [similar description for $\vec{B}$], $\vec{S}$ expresses the direction of energy flux density [loosely, the direction of ‘energy-flow’ carried by EM Wave. Whose time average provides the intensity of the wave $I = \langle |\vec{S}| \rangle \propto |\vec{E}_0|^2$], $\vec{k}$ expresses the propagation direction, also known as wave vector, which relates to velocity($\vec{v}$) or momentum ($\vec{p}$)(vectorially) and wavelength (scalarly)[$|\vec{p}| = \hbar|\vec{k}| = \frac{h}{\lambda}$]. Moreover, we have angular frequency (ω), frequency (ν), wavenumber ($\bar{\nu}$), and wavelength (λ) as other physical quantities, particularly associated with waves.

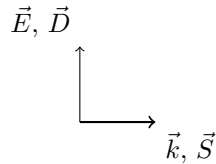
In short, all look like for free space $\vec{D} = \epsilon_0 \vec{E}$; $\vec{B} = \mu_0 \vec{H}$:



5. *Does Light also exhibit particle-like properties, and also has angular momentum?*
6. Yes.
7. *Can we further compactify the above figure?*
8. Yes. The magnitude-wise $|\vec{E}| \gg |\vec{H}|$, due to c (speed of light, universal constant and having a high positive value) in multiplication with $\vec{H}$. So, dominantly (grossly) light is a transversely propagating electric field $\vec{E}$, the direction of its oscillation always aligns with the direction of the polarisation vector $\vec{P} = \hat{E}$, by definition, irrespective of where the light is travelling through.

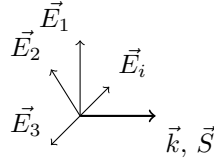
Our following description may be based on ignoring the $\vec{H}$ existence. Firstly, our whole idea is to understand the polarisation vector of the EM wave, its behaviour in different media, how it relates to optical properties, how optical phenomena are connected to it, etc.

In short,



9. *Are $\vec{E}$ and $\vec{k}$ always $\perp$, $\vec{E}$ and $\vec{S}$ always $\perp$, $\vec{S}$ and $\vec{k}$ always $\parallel$, and $\vec{D}$ is always along the $\vec{E}$ by definition $\vec{D} = \epsilon \vec{E}$, even in any media?*

10. Answer to all these is kind of **No**. Given situations hold in free space. We will see what happens to this vector arrangement in the media.
11. *Does the light just have one sinusoidally varying $\vec{P}$?*
12. **No**. We are convinced (by definition) that the direction of electric oscillation in the TEM wave is the direction of the polarisation vector of light. The such light we get from the Sun's fusion reaction is the result of charges oscillating randomly in such direction, and within such certain plane, resulting in many sinusoidally varying electric polarisation vector $\{\vec{E}_i\}$, but all propagating in a common direction $\vec{k}$, as single light ray. We call this ray a mixture of randomly oriented polarisation vectors, which is essentially an **unpolarized light**. In short,



13. *Then, how to filter out other electrical oscillations from unpolarised light?*
14. If one talks about the optical point of view of polarisation. An unpolarized ray may not be something of any needful to our discussion. So let's see how we can have polarised light. I know just two schemes for getting polarised light. Firstly, Upon reflection (on a regular plane surface), we get partly polarised light, because most randomly oriented polarisation vectors (mostly those having a significant part of the normal component of any polarisation vector, alining perpendicular to plane of surface) are just absorbed (or refracted or transmitted) in the surface, and thus we are left with mostly those polarisation vectors, which are kind of horizontal (or in a plane) to the surface plane. And if one keeps changing the angle of incidence such that the angle between the reflected ray and refracted ray is at normal, then I can assure you that your eyes will be receiving plane polarised light at incidence angle i_p , hence you removed the 'partly' by 'completely linearly'. Such an angle of incidence is called the **Brewster's angle** or polarising angle (i_p) (Fig:1).

This is not a robust method. So on thinking deeply, to attain the objective to kill(filter-out) the other randomly oriented $\{\vec{P}_{i-1}\}$, so that at the end, only one phasor [oscillating polarisation vector] survives. We make a call to the second scheme, whereby passing such a ray through a series of minutely gapped parallel slits (linear windows). In nature, we have a crystal, namely the tourmaline crystal (Fig:3), which has the same internal structure, and can be useful to get linearly polarised light, say along $\hat{z}$. Such a crystal here acted as a linear **polariser**, and we get $\hat{P}_z$ light, a linearly polarised light. If you bring another such polariser whose orientation (direction of gap) is random, we can analyse whether the first output is polarised light or not; thus, we call that an **analyser**. The output ray will be down in its intensity in comparison to the previous (un)polarised light. **Malus's law** says $I = I_0 \cos^2 \theta$, (Fig:2). This way can produce, importantly, vertically polarised rays and horizontally polarised rays.

Let's denote the state of such an electric field oscillation, aka polarisation state, respectively, as $|V\rangle$ and $|H\rangle$. It shows that polarisations are 'states', and should carry the information (why

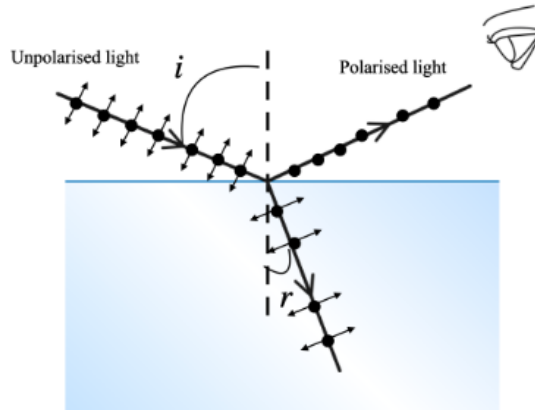


Figure 1: Reflection at polarising angle (Google)

such an association information state is unknown to me now, or may I need to think (read) more), which is what we deal with in photonic quantum computations. One can encode this as $(0; 1)$ and $(1; 0)$, hence these state an example of a qubit in a photonic system. This hints us towards the existence of quantum logic gates (unitary transformation that to be depicted via a complex matrix), other superpositioned photon states. We may later talk about these early “gut-feelings”.

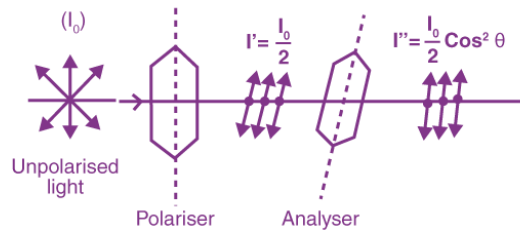


Figure 2: Malus's law (Google)

15. *Will there be only $|H\rangle$ and $|V\rangle$, and won't they obey the vector (phasor!) addition law, to give other superpositioned states?*
16. **No**, they can make more states, and **Yes**, we can visualise how a new state can be produced via the addition law, and even in a lab, it can be achieved via optical devices. If we need to know how to get the some standard superposition polarisation state (out of the basic two discussed states), aka $|D\rangle$: Diagonally polarised state of light, elliptically (both left and right) polarised light, or circularly polarised light, and can we represent them and their superposition state in a column(row) vector? And what set of optical devices in the lab, capable of generating these states, or should I say capable of performing a transformation (those may be unitary)?

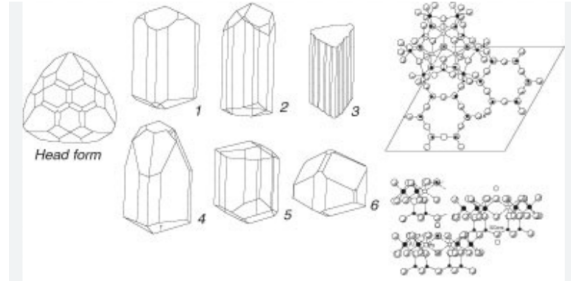


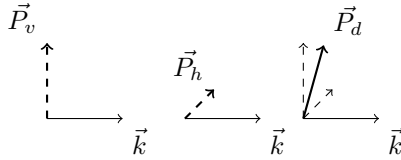
Figure 3: Tourmaline crystal (Google)

Let's take a deep dive, while keeping the wave picture in mind.

To have a diagonal polarised light, we need to understand that polarisations are vectors, and they should follow the vector law (isn't it?), thus if provided with atleast two EM waves, we should be able to add two polarisation vectors to get a resultant vector, which correspond to new TE wave in itself (but diagonally polarized), also the addition of waves can come up with lag or lead in phase. So, to get a diagonal polarised ray, we need to bring vertically and horizontally polarised waves together in perfect phase sync (may or may not be of same amplitude), since they are oriented orthogonally, the resultant sum of their phasors will be a new phasor, but in the diagonally orientation, thus we get a diagonal polarised wave. Now, this diagonal polarised ray is just a superposition of vertically and horizontally polarised light, and in state notation can be expressed as,

$$|D\rangle = \frac{1}{\sqrt{2}}(a|H\rangle + b|V\rangle)$$

Where $a^2 + b^2 = 1$.



17. *Can we make this happen in the lab? and is it only a diagonal polarisation possible?*
18. **Yes**, we can make it happen in the lab, and also, that's not the only superposition state; we have more standard superpositioned states. We should be using a crystal (as an optical device), which directly gives diagonally polarised light. One way is to consider either H- or V-polarised light, take any linear polariser, set it at ± 45 degrees relative to the given polarisation vector, and the output will be diagonally polarised light. Also, **half-wave plate (HWP)** can perform the same task, converting linearly polarised light to diagonal or anti-diagonal light.

If, while adding the phasor of two linear polarised waves with a certain relative phase lag (ahead) by say $\lambda/4$ (In lab, by optical device, the quarter wave plate), we can get circularly

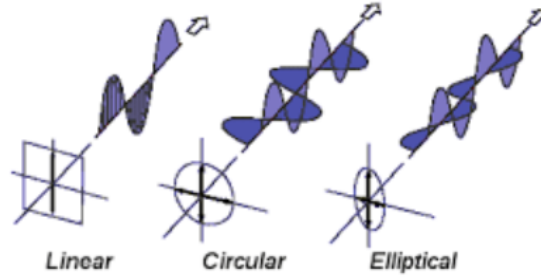


Figure 4: When an observer at the front sees the incoming light. (Google)

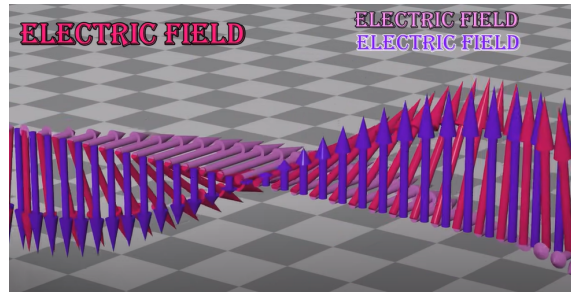


Figure 5: Red is circular polarised light.(Youtube)

polarised light or elliptically polarised light (Fig:4). As the name suggests that in circular polarisation, the propagation of the electric field oscillation seems like a phasor rotating in a circle, while in elliptical polarisation, it rotates tracing an ellipse when an observer sees incoming light. As if they are being put together, at a $\pi/2$ phase difference (introduced by optical component: QWP) but same amplitude, the resultant phasor depicts circular polarised, otherwise elliptical polarised (amplitudes unequal or phase is $\delta \neq \pm\pi/2$). To detect whether light is circularly or elliptically polarised, we find a constant intensity output in the case of circularly and no zero, noticeable variation in intensity in elliptically polarised light. Provided that the input was the output that was passed through the continuously varying analyser.

19. *Is Polariser, HWP, PBS, QWP only optical devices?*
20. No, there are more, like Prism, mirror, Lens, Optical fibre, Wave guide, PDBS, PBS, BS, Aperture, Modulator, Rotator, interferometer, Wave plates. Later, we may discuss them. It should be noted that, as we are calling polarisation states ‘information states’ of light, then these optical devices should be some kind of logic gates, which do a transformation of a given initial state to a final state, as if they are a unitary operator. Also, there is a set of detectors, which will collapse the superposition state and carry out the measurement, so they can be thought of as a projector.
21. *Why It looks like our discussion is converging towards polarisation, as being the central to the*

discussion?

22. If referring to a book [P.E. Powers, Nonlinear Optics], we can understand why polarisation is central to our discussion, via Poynting's theorem.

$$\oint \vec{S} \cdot d\vec{a} = - \left[\frac{dW_{mech}}{dt} + \frac{\partial}{\partial t} \int U dV + \int \vec{E} \cdot \frac{\partial \vec{P}}{\partial t} dV \right]$$

Where $\vec{S}$: it is instantaneous power flow per unit area., U is electromagnetic energy density in the fields, W_{mech} : mechanical work done on charges, and the third term is the one that matters for *light-matter interaction*.

This generally means that if the Laser enters the small volume of a medium, it will come out with the lesser energy, because energy exchange between the electric field and the medium has occurred.

The reason might be: *Polarisation is central to energy exchange between a medium and an electric field. The polarisation of a medium plays a key role in the energy dynamics of a system, and is also central in the study of nonlinear optics*¹. And, the magnitude of the energy exchanged is proportional to the time average of $\langle \vec{E} \cdot \frac{d\vec{P}}{dt} \rangle$, thus, it is an energy flow term.

(just an early supposition!) We can also recall, whenever we talk about the existence of 'phase relationship' between the field of polarisation, in some way we are referring to this 3rd expression, and also, it might be related to the word 'parametric or cause of 'phase matching'.

23. *Isn't the elementary understanding of polarisation come from the electric dipole? If yes, then how does polarisation connect with the ongoing discussion? In short, what is the microscopic point of view of polarisation?*
24. When we hear about polarisation (specifically the dielectric polarisation) $\vec{P}$, maybe for the first time, in electrostatics, it generally comes through the discussion one makes about free ($_f$) and bound ($_b$) charge densities (ρ) and currents ($\vec{J}$), (ρ_f , ρ_b , $\vec{J}_f$, $\vec{J}_b$). Sometimes in literature, the term *free* is connected to *external*, and the term *bound* is connected to *induced*. And these physical quantities are highly related to microscopic electric dipoles in the media (may be isotropic or anisotropic, each while showing linear or nonlinear behaviour). In formal terms, when we subject the media to the external electric field, the media gets induced and a separation of charges occurs, creating small electric dipoles, and all dipoles would try to align along the direction of electric field, we call that generated (induced electric field as polarisation vector), and say the media is polarised. Thus, referring to a book [KK Sharma, Optics],

$$\vec{P} = \frac{\text{net electric dipole moment in a macroscopically small volume } V}{\text{Volume } V}$$

And,

$$-\nabla \cdot \vec{P} = \rho_b$$

¹From the book.

and similarly dynamical quantity,

$$\vec{J}_b = \frac{d}{dt} \vec{P}$$

I was motivated to notice that one should be clubbing them under a continuity equation (σ is here the current source-sink term (exists due to binary electric poles), for conserved electric charge $\sigma = 0$),

$$\nabla \cdot \vec{J} + \frac{d\rho}{dt} = \sigma$$

Where, $\vec{J} \stackrel{?}{=} \vec{J}_f + \vec{J}_b$ and $\rho = \rho_f + \rho_b$.

Not very important for ongoing discussion, but still, if one is interested in considering the magnetisation $\vec{M}$ in the media as well, then,

$$\vec{J}_b = \frac{d}{dt} \vec{P} + \nabla \times \vec{M}$$

We can construct the relation between the external electric field, the induced polarisation vector and the electric displacement vector via the following **constitutive relation**(see²):

$$\vec{D} = \epsilon \vec{E} + \vec{P}$$

Now shifting our focus, note that $\vec{D}$ is an electric displacement field or electric induction, and $\vec{E}$ is the electric field. We have electric polarisation density known as a polarisation $\vec{P}$. The physical fields $\vec{E}$ are connected to respective induction parameters $\vec{D}$ through their respective multiplicative constants; for the electric case, it is the permittivity, ϵ . The physical fields $\vec{E}$ are connected to respective polarisation densities $\vec{P}$ by respective other constants. For the electric case, it is the electric susceptibility χ times ϵ_0 , that is $\vec{P} = \epsilon_0 \chi : \vec{E}$, here $:$ is tensor product, as $\chi = \chi_i^{(1)} + \chi_{ij}^{(2)} E_j + \chi_{ijk}^{(2)} E_j E_k + \dots$. Remember a beautiful relation for the electric case

$$\epsilon = \epsilon_r \epsilon_0 = (1 + \chi) \epsilon_0$$

25. *Can we explain Refractive index and dispersion? (Isn't that violate our initial constraint that we will first talk about light in free space, then we move to light in media?)*
26. Yes, Yes. By recalling, another useful relation in this chapter is

$$n = \frac{c}{v} = \sqrt{\epsilon_r \mu_r} \approx \sqrt{\epsilon_r}$$

As $\mu_r \approx 1$ for most crystals (non-magnetic optical materials). So whenever one talks about the Refractive index, one is basically hinting towards either phase velocity or electric permittivity or electric susceptibility, and these are being scalar or tensor, just depending on the media type being isotropic or anisotropic. Thus, from an advantageous point of view, in a

²It seems like $F_{field} = I_{induced} - P_{polarised}$ or physical Fields is an Induction field minus its Polarisation density.

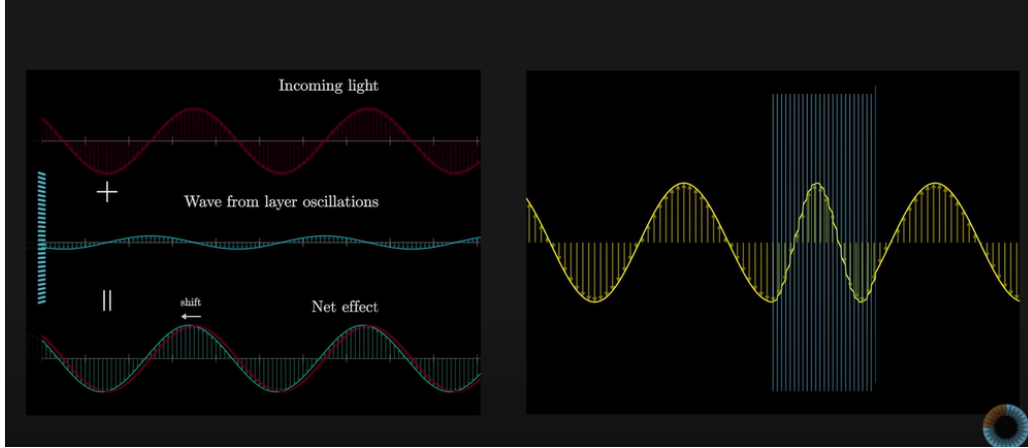


Figure 6: Mechanism showing how light slows inside a medium. (Youtube)

homogeneous, lossless medium, the **Refractive Index** (R.I.) is a direct way to talk about the relative permittivity tensor ϵ_r without diving into atomic details. It determines the phase velocity, $v_p = c/n$, which is why a better understanding of R.I. is expressed in standard literature through the terminology involving phase velocity, since phase velocity controls how the wavefront advances inside the medium. Still, we keep our discussion mainly centred on the refractive index. For example, to understand phenomena like **normal dispersion**, it can be expressed as $n(\omega_i)$, which means that light of fixed frequency, when passed through the media, feels that the material is offering n_i refractive index. In short, dispersion means the sensitivity or dependence of the refractive index on a given frequency of light travelling through the medium. In actuality, higher frequency waves travel with lower phase velocity (experience higher separation (dispersion) from the tentative path) compared to lower frequency waves; this separation is due to optical density, whose physical measure is refractive index n , and from a microscopic light matter interaction point of view, described by the relative permittivity tensor.

It also raises several questions: why the light gets slow, why crystals offer resistance, such things can only be satisfactorily understood when we look within the crystal and how, if making light $\vec{P}$ and $\vec{k}$ or any other to changes. That will be done later.

27. *But why does wave travel slower or faster?*
28. Due to the Phase kickback or phase kickforward, which occurs when the light passes the interface of two media. A medium can be understood as composed of several layers. As the light passes through any layer, the layer oscillates (Due light-matter interaction occurrence: the layer oscillates because of incoming light is sinusoidally varying and the layer's dipoles try to align in the direction of light's oscillating electric field) as a result, a wave is generated with a small amplitude, at the same frequency but not in phase, due to layer wiggling (oscillates). This wave interferes with the incoming light wave, resulting in an overall phase kickback. and also makes it clear why RI is generally greater than 1. (Fig:6).
29. *Just like we saw the list of physical quantities, is RI also a physical quantity in the same sense,*

and also, under what group should we keep dispersion?

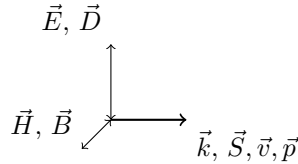
30. I will refrain from directly calling RI a physical quantity in the same sense as those listed quantities say $\vec{k}$ or $\vec{D}$, etc. In fact, a better technical common term would be ‘optical property’. And dispersion is an example of an optical phenomenon. I am giving a relevant list of optical properties and optical phenomena.

Optical properties	Optical phenomena
Electric Susceptibility	Dispersion
Permittivity	Birefringence
Refractive Index	Nonlinear Effects
	Interference
	Reflection
	Total Internal Reflection
	Abberation
	Refraction
	Difraction
	Conic Refraction

31. We conceptualised the dispersion just via the mathematical function $n(\omega_i)$. Can all other optical phenomena, similarly, be conceptualised as $f(x_i, y_i, \dots), g(x_i, z_i, \dots), \dots$. And it seems there RI played a crucial role. Will it do so in other optical phenomena as well, or do we need to invoke other optical properties instead of RI, such that they are dependent on optical physical quantities?
32. RI seems central to media, as crucial as the concept of polarisation is for the case of light. I think other optical phenomena, such as birefringence, can be compactified as $n(\theta, \hat{p})$, which means that the RI experienced by light inside a certain medium is direction and polarisation dependent. Thus, in both the above optical phenomena, we notice they are basically related to RI, and how RI is a function of single or multiple physical parameters. So, indeed $f(x_i, y_i, \dots)$ makes sense, provided $f \equiv n$. It would be interesting to come up with other optical properties rather than RI, then that would be $g(x_i, z_i, \dots)$. As of now, I don’t know $g \equiv ?$.

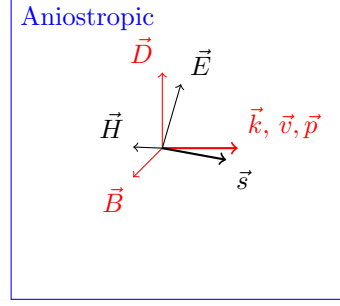
0.2 Within the material medium

1. Now it feels like we should investigate completely, to understand how light behaves in the media. The first question should be, does the light retain the following vector arrangement shown below, as it has in the vacuum, even when it enters the media?



If not, why do modifications occur, and what would be the new arrangement of vectors of light?

2. Answer depends on the type of media one is considering. A straight answer would be **No**, but there exist a class of media where given arrangements indeed remain intact (in isotropic media). But, in general (anisotropic media), the figure would change to:



Where, if $:$ denotes tensor product, then we say that a few parameters μ and ϵ are linked to medium properties (why?). We see, few splits as $\vec{D} = \epsilon : \vec{E} \implies \vec{D} \angle \vec{E}$ and $\vec{B} = \mu : \vec{H} \implies \vec{B} \angle \vec{H}$, but, still new set of mutually orthogonal physical quantities are given by $\vec{k} \cdot \vec{D} = 0$; $\vec{k} \cdot \vec{B} = 0$; $\vec{B} = (\dots) \vec{k} \times \vec{E}$; $\vec{D} = -(\dots) \vec{k} \times \vec{H}$; and finally, the Poynting vector splits from propagation vector, $\vec{S} = (\dots) \vec{E} \times \vec{B} \implies \vec{S} \angle \vec{k}$, (this angle is referred as Walk-off angle) (Why?). In summary, $\vec{E} \angle \vec{D}$, $\vec{D} \perp \vec{k}$ and $\vec{s} \parallel \vec{k}$ so both $\vec{E} \angle \vec{s}$ and $\vec{D} \angle \vec{k}$, finally $\vec{E}(\angle)(\perp)\vec{S}$. I am not sure whether all these four vectors are coplanar or not, but I still go with they being coplanar.

3. What are the types then, and how is a parameter, say ϵ linked to medium properties?
4. There exist two classes of media. isotropic and anisotropic.

An **isotropic** crystal (eg: glass, most metals) has a refractive index that does not vary depending upon the direction in which the electric field is oscillating (aka polarisation vector) of incoming light.

An **anisotropic** crystal (eg: glass, most metals) has a refractive index that varies depending upon the direction in which the electric field is oscillating (aka polarisation vector) of incoming light. Now inside the anisotropic media, since RI varies, it must have a minimum value and a maximum value; the difference of RI between these two values is mathematically called **Birefringence**, $\Delta n = n_e - n_o$. We will later show this is also an optical property, but part of an optical phenomenon called double-refraction or Birefringence.

And, note from the relation, $\epsilon = \epsilon_0(1 + \chi)$, so this electric susceptibility tensor χ is basically a measure of the electric properties of the medium. But more deeply, it is further connected to refractive index n (an optical property) when ignoring $\mu_r \approx 1$ (in case of media possessing non-magnetic properties as well)

$$(\chi + 1)^{1/2} \approx n$$

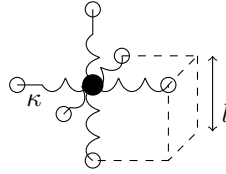
On taking the binomial series expansion,

$$1 + a\chi + b\chi^2 + c\chi^3 \dots \approx n$$

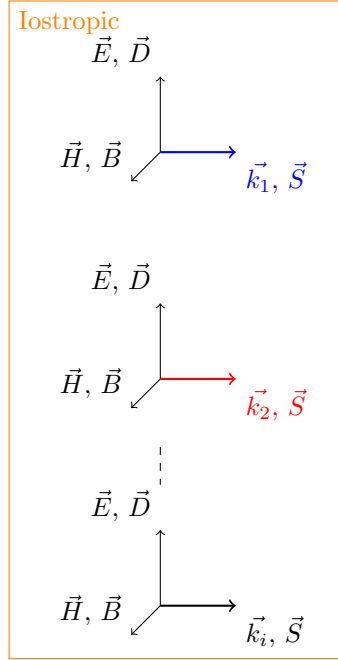
5. *What may be the logic behind the expansion of $(\chi + 1)^{1/2}$?*
6. By expansion, we can explain when the medium would behave linearly and non-linearly. In fact, for an order of χ more equal to 2, the media would exhibit a non-linear response. and if χ is zeroth or 1st order, then the media is referred to as linear. It is important to notice that anisotropy is something independent of whether one is linear or nonlinear. We will later see what makes a medium anisotropic.
7. *What makes media anisotropic, isotropic and vacuum?*
8. When talking about crystal, we learned that there exist 7 crystal systems, varied in crystal parameters (edge length: a, b, c and edge angle: α, β, γ).

If having a cubic lattice $a = b = c = l$ and $\alpha = \beta = \gamma = 90^\circ$, then such a isometric crystal (a term that expresses a geometric point of view) is called an isotropic medium. The term ‘tropic’ expresses the media’s property in direction.

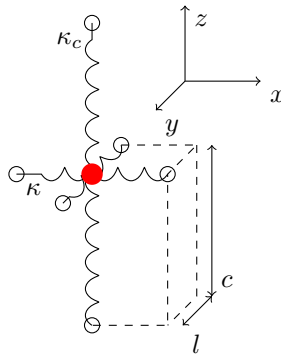
Already stated examples are glass, metal, etc. Further, we can model the vertex by a point mass and the edge as a spring as follows. In the case of isotropic media, the stiffness constant κ is fixed along any edge length. Here, we consider a site (black) and show how uniformly it is connected to all other nodes in 3D space.



Now, when light passes through this isotropic medium, optical properties would remain the same in any direction, since in any direction, when light interacts with atoms or charges, they will execute harmonic oscillations around the mean, and respective harmonic oscillation has the same magnitude in any direction, since the stiffness parameter is constant. But only if the light is non-monochromatic light or a mixture of multiple wavelengths, then dispersion will occur (note $|\vec{k}|$ related to λ , and dispersion is $n(\lambda_i)$):



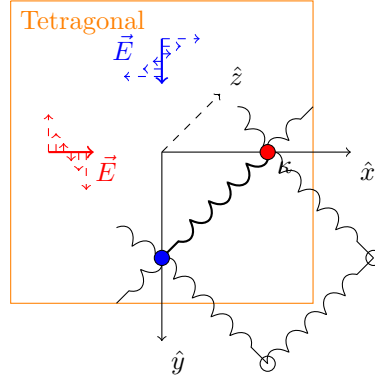
In a similar setting, we can model an anisotropic medium (in fact, all six out of seven crystal structures are anisotropic). Consider the example tetragonal unit cell ($a = b \equiv l \neq c$ and $\alpha = \beta = \gamma = 90^\circ$), and there should be two stiffness parameters, κ and κ_c . Now, when light passes through the anisotropic medium, the optical properties would not remain the same in any direction, since in some directions the stiffness parameters are different. For example, to wiggle the red atom (say) of a tetragonal crystal, along the vertical direction is harder than in the horizontal plane (if $\kappa_c > \kappa$). So if a light has an oscillating electric field $\vec{P}$ in the same plane as the crystal's horizontal plane (x-y plane), it experiences the same optical properties within the plane from any direction (2D), but different properties appear when P has a component perpendicular to the horizontal plane.



It may seem that within an anisotropic crystal, we have stacks of consecutive numbers of planes (x-y), within which light exhibits optical properties equivalent to as if it were in a

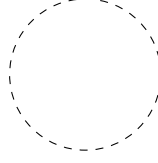
isotropic media, the ray is termed an **ordinary ray**, and the direction of the stacking is termed the **optical axis**. Since tetragonal has one optical axis, it is an example of an anisotropic **uniaxial** crystal.

Let's draw the uniaxial crystal (edge length $a = b \equiv l$ units, or horizontal cross-section), and see (the top view) how light travels from any direction $\hat{x}$, $\hat{y}$, and a linear combination of these two bases, behaves. Note that in all cases, the sinusoidal oscillating polarisation vector $\vec{E}$ is within the basal plane $\hat{x}\hat{y}$. Thus, making propagation vector $\vec{k}$ (red and blue arrows) always perpendicular to optical axis, $\hat{z}$ (out of the page). If there is an mass in the line of $\vec{k}$, it will wiggle (Hooke' law, with low amplitude, due to finite κ value), irrespective of light approaching it from any direction in the plane; the wiggling will be the same, because the stiffness parameter κ is the same; generating wiggling wave, which interfere with incoming light, and resulting a phase shifted light, which appears slowly moving (in comparison to light in free space), which we call is happening in the crystal, due to $n > 1$ value of refractive index, thus refractive index is same in any direction within the plane of uniaxial crystal. And if one agrees with the fact that the refractive index experienced by O-ray is the same in any direction, then ultimately it agrees with the fact that the velocity of O-ray is the same in every direction (within the xy plane). A piece of note, that blue, red or even on the next page, the black linearly polarised light is just for depiction and has nothing to do with the colour of light. (I just wanted to make them look different, so I colored them differently)



Therefore, light behaviour is the same along any direction, in the plane of a uniaxial crystal or the plane perpendicular to the optical axis of a uniaxial anisotropic crystal. So basically, things appear as if it is isotropic but confined in 2D, so light is just behaving ordinarily as it behaves in an isotropic crystal, therefore classified as an ordinary ray. Thus, I see no conceptual gap in saying this refractive index is the same in all directions within the plane. I think one can explicitly show how this stiffness parameter and RI are mathematically linked.

And I refrain from modelling vacuum (free space) by a set of infinite harmonic oscillators, so we just denote it by a sphere.



So it is clear that light in isotropic media would retain the same arrangements of physical parameters as it does in free space, but only the new thing it experiences will be an optical phenomenon, dispersion $n(\omega_i)$. It can be said that isotropic media show spherical symmetry. And, inside the anisotropic media, we may at least find a family of parallel planes, wherein light behaves ordinarily, where such light will be travelling with a polarisation vector oscillating within such a plane.

We will later discuss what would happen to the light if its polarisation vector has components along the optical axis of anisotropic media.

9. *Can it be shown that stiffness κ (crystal's mechanical property) and n (crystal's optical property) are mathematically linked, either through a direct relation or an indirect relation?*
10. Let's refer to the book, Nonlinear optics by Powers.

Cosidering

$$10.1. \text{ Hooke's law: } \bar{F}_r = -\kappa d = -m\omega^2 \bar{X}$$

$$10.2. \text{ Damping Force: } \bar{F}_d = -m\Gamma \dot{\bar{X}}$$

$$10.3. \text{ Electron's electric force: } \bar{F}_e = -e\bar{E}(t)$$

Where, $\bar{E} = \frac{1}{2}Ae^{-i\omega t} + c.c.$, here, E is the time-dependent oscillating (at ω) electric fields of light (electric polarisation vector), A is the complex amplitude of incoming light. So, when light hits the mass, there will be say 'linear' response by the crystal, and the oscillator will oscillate at the same frequency (considering an ideal situation) as the driving force, but out of phase, the displacement is $\bar{X} = \frac{1}{2}xe^{-i\omega t} + c.c.$, Here again, x is the phase shift of linear displacement, a complex amplitude like A . So the equation of motion, from Newton's law,

$$m\ddot{\bar{X}} = -m\Gamma \dot{\bar{X}} - m\omega^2 \bar{X} - e\bar{E}$$

Upon solving the Equation of motion, after substituting the form of $\bar{E}$ and $\bar{X}$. We see,

$$x = -\frac{eA}{mD(\omega)}$$

Where, $D(\omega) = \omega_o - \omega^2 - i\omega\Gamma$ From understanding of polarisation (N number of dipoles), we know polarisation's complex amplitude is related to this phase shift in displacement via

$$P = -Nex$$

On substituting the x , we get

$$P = Ne^2 \frac{A}{mD(\omega)}$$

and comparing with $P = \epsilon_0 \tilde{\chi}^{(1)} A$, We see a complex number, first-order linear electric susceptibility as,

$$\tilde{\chi}^{(1)} = \frac{Ne^2}{\epsilon_0 mD(\omega)}$$

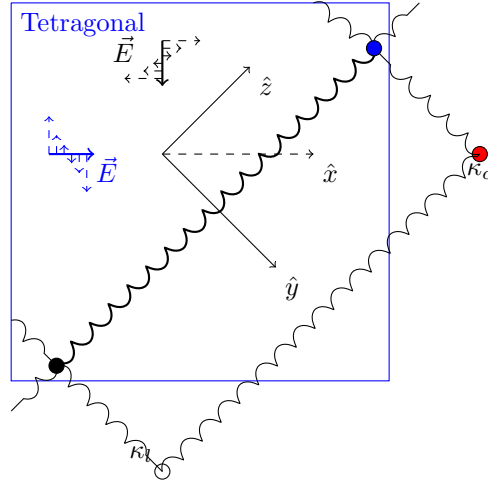
From the study of electromagnetism, we now, there exists a relation between the real part of electric susceptibility and the refractive index as follows:

$$n^2 = 1 + \chi^{(1)}$$

Now, having done the above, we know that $\bar{X}$ is related to stiffness κ via Hooke's law, which is related to χ (through the presence of $mD(\omega)$) and that is further linked to refractive index as shown above. This, I think, provided a satisfactory mathematical argument for our visualisation and claim. And, I must say the relation is not explicit but *indirect* in some sense. So,

$$\kappa \propto n$$

11. *Considering if light ($\vec{E}$) moves in an anisotropic crystal, but not within a uniaxial plane, but across it, at arbitrary inclination, then?*
12. Physicists named such a ray of light as an **extraordinary** ray. Let's understand what 'extra' is in the extraordinary ray. Let me consider the same example of anisotropic media, but a vertical cross-section of a tetragonal crystal.



It is clear from the diagram that if light imparts energy to either mass pairs (blue-black or blue-red), firstly, both mass-spring pairs will oscillate differently (upon comparison), and

secondly, the amount of wiggling depends on whether black-polarised light or blue-polarised light was the cause, because the angle of incidence is different in both cases. I can say, if provided $\kappa_c > \kappa_l$, then the lower wiggling amplitude will be associated with the blue-black pair than the blue-red pair. So the refractive index experienced by incoming light due to the blue-black pair will be different (larger) from the blue-red pair. In our case, both black-polarised and blue-polarised light are examples of extraordinary rays, and both behave differently in themselves, because the polarisation vector makes a different angle with the optical axis. Note, the polarisation vector in both cases does not lie within the xy plane.

Now, imagine merging both the cross-sectional diagrams of tetragonal, revealing that it supports the propagation of two kinds of rays; one is the o-ray and the other is the e-ray. Any two o-rays are equivalent, but any two e-rays aren't, because of incident angle dependence, or e-rays are sensitive to direction, and thus, correspondingly experience respective values of RI, and thus should be travelling within a velocity range, such that it may be possible that e-ray may travel with velocity either greater, or lesser or even equal to ordinary rays inside the tetragonal anisotropic uniaxial crystal. So the 'extra' in the e-ray is just its direction dependence RI (an optical property), apparent due to the physical quantity $v_{re}(\theta)$.

We have probably understood that RI is just a superficial property, and thus at the core it is the velocity (in fact phase velocity) which is a real optical physical quantity, and from the lecture or book on linear optics, it can be understood that the o-ray follows:

$$v_{ro} = \frac{c}{n_o}$$

This is what we call in standard literature Snell's law. And, e-ray obeys,

$$\frac{1}{v_{re}^2} = \frac{\sin^2 \theta}{\left(\frac{c}{n_e}\right)^2} + \frac{\cos^2 \theta}{\left(\frac{c}{n_o}\right)^2}$$

Where v_{ro} denotes the phase velocity of the o-ray, v_{re} denotes the phase velocity of the e-ray, n_e is the refractive index of the e-ray, n_o RI for the o-ray, and θ is the angle the ray makes with the optical axis (that $\vec{k} \angle \hat{z}$).

Now, from the above expression, it can be seen that if $\theta = 0$, then $v_{re} = v_{ro} = \frac{c}{n_o}$, so both the rays would have the same speed (when the e-ray propagates $\parallel$ to the optical axis). See (Fig:7). And if the second term vanishes, provided $\theta = \pi/2$, then $v_{re} = \frac{c}{n_e} \neq n_{ro} = \frac{c}{n_o}$, this will introduced either a lag or lead, which will further depend, whether crystal is **positive uniaxial** ($n_e > n_o$) or **negative uniaxial** ($n_e < n_o$). Which I think is somehow related to the kind of relation we have with κ_c and κ_l , and how these relate to RIs. In the general case, we can claim that $n_e \neq n_o$, so it is better to define $\Delta n = n_e - n_o$, and literature calls this value **birefringence**, so a negative Δn makes us call the crystal negative uniaxial and vice versa.

13. *How to convince that when the incident is exactly along the optical axis, then both e-ray and o-ray will not separate but remain as a single ray?*
14. See Fig:8, and notice that both horizontal and vertical polarised light experiences the same RI, since the value of the stiffness parameter κ_l is the same for both.

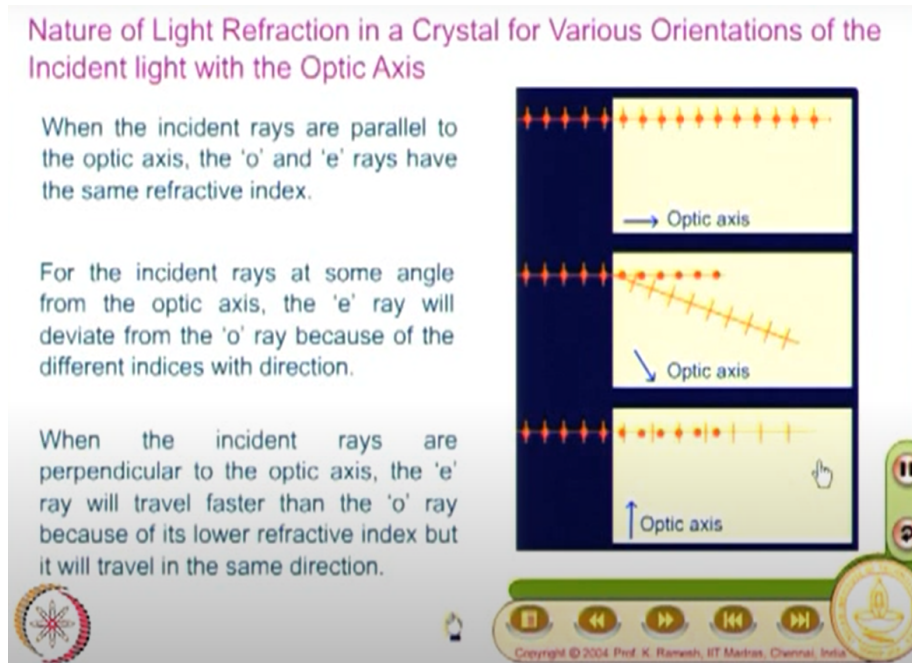


Figure 7: A SS of simulation, showing how unpolarized light behaves when entering anisotropic media, the optical axis is oriented differently. (Youtube)

I am a bit doubtful at this stage, it seems to me that “to imagine” the combined case, mandatory makes or requires my incoming light to always be unpolarised (?) Can it be linearly polarised, or may it be just an o-ray or e-ray? So far in lectures, the unpolarized light case is considered, but I have read cases where $e \rightarrow o + o$, $e \rightarrow e + o$, etc, are taken to achieve ‘something’. Now, how should I be interconnecting these?

15. Can we, with a picture, understand the lag or lead of e-ray, when light makes an angle $\pi/2$ with the optical axis?
16. I have sketched a picture (see Fig.9), which may resolve the concern. Based on an intuitive picture, it shows, vertically polarised light is not lagging, because it may weakly interact with vertical springs. As $\kappa_c > \kappa_l$, thus lower phase kick back and moves with faster velocity, experiencing lower refractive index. $\Rightarrow n_e < n_o \Rightarrow \Delta n < 0$, so not only does it experience by the e-ray, since its polarisation vector is not lie within the uiaxial basal plane or propagation vector $\vec{k}$ is perpendicular to the optical axis, but also it expresses that the crystal is a negative uniaxial crystal.
17. Before completely answering the ongoing question, should we formally try to express the optical phenomenon of **double refraction**?
18. Yes. Let me put it this way: Birefringence, or double refraction, is the separation of light into two polarised lights inside a crystal, one being the ordinary ray and the other the ex-

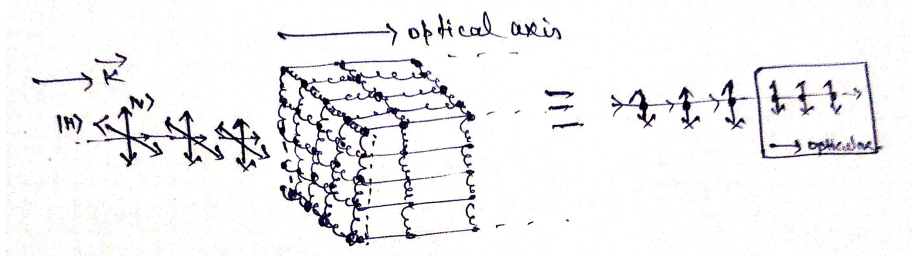


Figure 8: A case in the uniaxial crystal, if $\theta = 0$.

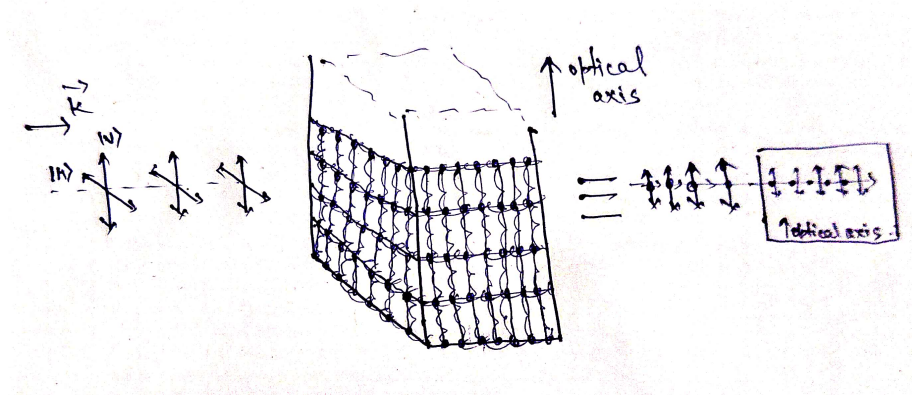


Figure 9: A case in a uniaxial crystal, if $\theta = \pi/2$.

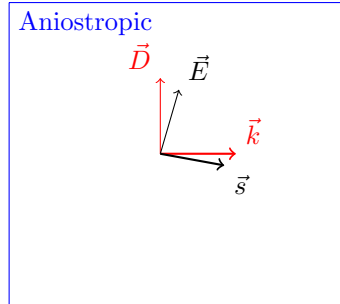
traordinary ray. The first has a plane of polarisation perpendicular to the optical axis, and the latter has the plane of polarisation (with a component) along the optical axis. In short, $n(\theta, \vec{k}_i)$. It is nice to recall how this is different from normal dispersion, which is $n(\omega_i)$. So it seems that light inside the crystal is generally affected by its energy parameter $\hbar\omega_i$, under the name of normal dispersion and momentum parameter $\hbar\vec{k}_i$, which is reflected as the phenomenon of double refraction. Further, the respective phenomenon is either characterised by frequency/wavelength or polarisation of light. You should be asking why we didn't write $n(\vec{P})$ but $n(\vec{k})$, (while depicting birefringence) isn't it what we are talking about, polarisation ($\vec{P}$) instead of propagation $\vec{k}$ direction? Yes.

To answer the query, we note that $\vec{D} \cdot \vec{k} = 0$ and $\vec{D} \angle (\vec{E} \equiv \vec{P})$, so just via these two, it is clear that,

$$\vec{k} \cdot (\epsilon : \vec{P}) = 0$$

So encoding the information of $\vec{k}$ ultimately incorporates the information of $\vec{P}$, and in fact the angle between $\vec{k}$ and $\vec{s}$, say if ρ , then the same angle exists between $\vec{D}$ and $\vec{E}$. This makes

an angle between $\vec{E}$ and $\vec{k}$ equal to $90^\circ - \rho$. Or in words, the angle between the electric polarisation direction and the propagation vector should be $90^\circ - \rho$. This further motivates us to express $\vec{E} = \vec{E}_\perp + \vec{E}_\parallel$; these resolutions of $\vec{E}$ are considered with respect the direction of propagation vector $\vec{k}$. On a similar line of thought, I see no exaggeration in saying that $\vec{s} = \vec{s}_\perp + \vec{s}_\parallel$ should also be a valid result.



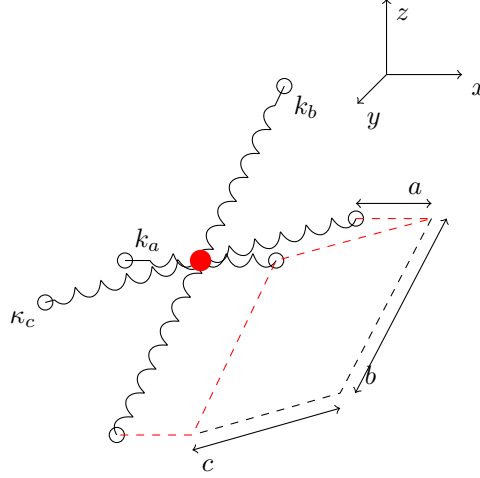
The use of such vector resolutions may be utilised in a later section to make a point.

19. *How to understand angle ρ ?*
20. We know the name, it is called **Walk-off angle**, which means e-ray, propagating in direction $\vec{k}$ has its Poynting vector not in parallel to $\vec{k}$ but makes angle ρ , seems like $\vec{s}$ walk-off from $\vec{k}$, inside anisotropic media. It can be intuitively imagined as if e-ray depicts a drunk person, who, due to dizziness, wants to go to $\vec{s}$ direction but ends up going to $\vec{k}$ direction (based on the lecture discussion). So, a Walk-off is an optical phenomenon, which is part of an e-ray understanding, and falls under the birefringence optical phenomenon.
21. *Will a walk-off happen when e-ray propagates along the optical axis?*
22. The answer found is No. The reasoning is simple: e-ray has nothing in its way, which allows this walk-off to occur, as e-ray behaves ordinarily when propagating in the direction of the optical axis. Both o-ray and e-ray move with the same velocity and thus experience the same refractive index, and finally both have merged $\vec{s}$ and $\vec{k}$. Hence, no walk-off.
23. *I have heard about fast and slow axes, how do these get incorporated into the above discussion?*
24. I don't know. I can guess based on definitions, when I refer book, Optical quantum computation, A. Pathak, I came across this detail, which boils down to: If $n_e < n_o$, then the extraordinary axis is the **fast axis** and the ordinary axis is the **slow axis**, and the terminology gets switched, based on the type of uniaxial crystal is one having.

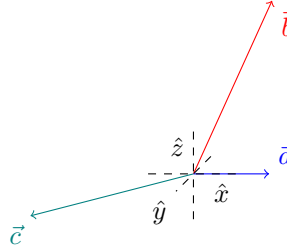
Based on whatever we asked and tried to answer (particularly the question related to the $\theta = \pi/2$ case), indeed support this notion. Since an ordinary ray will intersect more heavily with κ_l , thus imparting more energy to masses, and it will experience a greater refractive index (optical density), and thus move slowly, so we may call this moving (...) slow axis. I still have a gap in visualisation, as in place of (...), should I write parallel or perpendicular?

25. *Saying $n(\omega, \theta, \vec{k})$ is valid?*
26. I think, it should be valid to say so, because when light enters the anisotropic media, it should be going through multiple optical phenomena simultaneously, like normal dispersion, and double refraction, reflection, etc, if provided with the necessary conditions or ingredients for them to occur. It has also answered the supposition that we had regarding whether any f exists, such that $f(x_i, y_i, \dots)$. Here $f = n$.
27. *What about $g(x_i, z_i, \dots)$?*
28. I don't know yet. So far, everything seems to be getting packed under n . If any g exists, then it should be distinct from n .
29. *Can we summarise the advantages of isotropic media and anisotropic media?*
30. I read in book [Sommerfeld, Optics], that put, "The crystal structure of an anisotropic medium permits two waves with different linear polarisations and different velocities to propagate in any directions." And this was absent in the case of an isotropic medium. In fact, "In an isotropic medium, two linearly polarised oscillations in a crystal degenerate into elliptically polarised oscillations", (because as we discussed, elliptical polarization is just a superposition of two mutually orthogonal linearly polarised states, having phase difference).
31. *Can we depict all the discussion on uniaxial, through a diagram showing the arrangement of vectors of respective kinds of rays?*
32. Yes, but I may not do it now.
33. *We are convinced that the value of the refractive index is a function of θ , or just say, sensitive to direction. Since we live in 3D space, and we work with 3D crystals, can we be motivated to choose just three directions, and should we discuss subsequent follow-ups of this consideration?*
34. Yes, and we should. The very realisation that we live in 3D space and work with 3D crystal allows us to mark at least three directions within the crystal. Let's designate these directions by the term **principal axes**. Next, if one finds the value of RI at those special directions, one gets n_z , n_x , n_y , respectively. It is important to note that, even before the designation of the principal axes, the optical axis exists in the crystal, due to structural reasons (already discussed before).

Let's consider the most general case, which motivates us to investigate, anisotropic crystal. A triclinic crystal unit ($a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma$) is an example of such a type.



It will make things easy if we show the unit cell's edge length vectors. Let's make a figure. At the vertex, in addition, I put three mutually orthogonal lines for reference purposes (it is needed, because in a triclinic lattice, all angles are unequal).



I know this is not the standard way of introducing the concept of principal axes, still, bear for a while. In standard settings, as per book (PE Power and KK Sharma); we start from the tensor form of ϵ , and claim that by a certain geometrical cut in crystal, all cross-terms of ϵ tensor (depicted as 3×3 matrix, may be beacuse, crystal is 3D) can be made zero (or matrix can be digonalised), and remaining values are just n_x , n_y and n_z , and to get this, we utilised connection between ϵ and n , and that was also already been discussed.

And, they also make use of the following expression (in terms of components of the electric displacement vector $\vec{D}$, energy density u and electric permittivity tensor ϵ), which resembles an equation of an ellipsoid,

$$\frac{D_x^2}{u\epsilon_x} + \frac{D_y^2}{u\epsilon_y} + \frac{D_z^2}{u\epsilon_z} = 1$$

Where further $\vec{D} = \epsilon_0 \begin{bmatrix} n_x^2 & 0 & 0 \\ 0 & n_y^2 & 0 \\ 0 & 0 & n_z^2 \end{bmatrix} \vec{E}$ related to electric field (or $\vec{E}$ depicts incoming light's magnitude and direction of oscillation, recall polarised state $\hat{E} \equiv \hat{P}$), also, because permit-

tivity ϵ and refractive index n are interconnected quantities via $n_i^2 \equiv \epsilon_{ii}/\epsilon_0$. Using all these relations, one can re-express the equation of an ellipsoid as follows,

$$\frac{x^2}{n_x^2} + \frac{y^2}{n_y^2} + \frac{z^2}{n_z^2} = 1$$

Where, $x^2 = \mu_r D_x^2 / 2u\epsilon_0$, similary for y^2 and z^2 . When we compare the above equation with the standard equation of an ellipsoid, one notices that n_x , n_y and n_z are the lengths of the semi-axes of the ellipsoid and in fact are called principal axes of the index ellipsoid. One of the most crucial information I received from the above equation is that the principal axes are mutually orthogonal, and it was made possible when we cut the crystal nicely.

Having this much information, I can indeed draw a general ellipsoid (see Fig. 10).

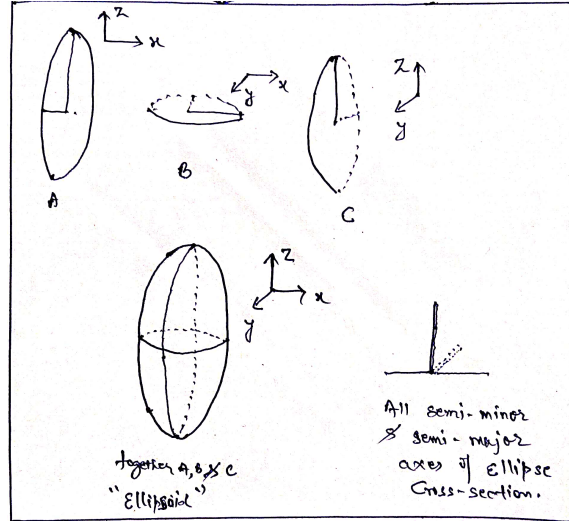
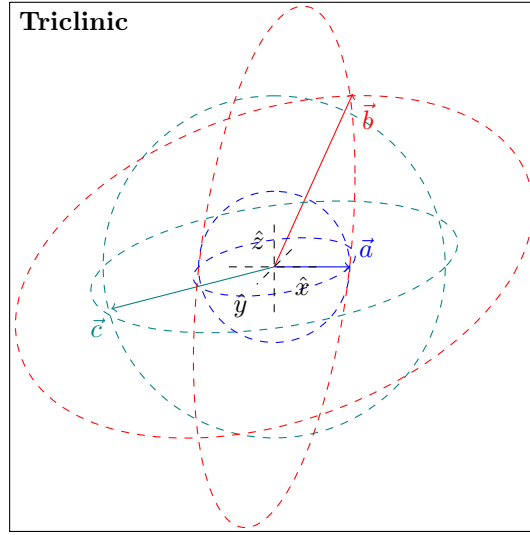


Figure 10: A Triaxial ellipsoid ($i = j \neq k$), other than this is just what we call an ellipsoid, is in fact a spheroid ($i = j \neq k$) or sphere ($i = j = k$). It took me a day to realise that this general (triaxial) ellipsoid **cannot** be generated by rotation about any “axis”, whether it is principal or optical, semi-axis (i, j, k), or just an arbitrary axis. Let me clarify that this sketch, which may have nothing to do with a tricline crystal, but just aims to show how one should think (visualise) about a Triaxial ellipsoid, if the equation is $\frac{x^2}{i^2} + \frac{y^2}{j^2} + \frac{z^2}{k^2} = 1$.

What should I do now with the triclinic thing? My mind-I, says, what if we consider triclinic edge vectors ($|\vec{a}|, |\vec{b}|, |\vec{c}|$) as the radius of respective circles, then we can have in totality three spheres (corresponding to each edge vector, red, blue and teal colored) that can be generated. Let's draw them,



Mind-II says what if we consider these edge vectors as if they are kind of “semi-axes” (non-perpendicular) for an oriented triaxial ellipsoid. By oriented, I meant that those semi-axes would be inclined at different angles. Let’s draw this version as well (Fig.11). Here, cross-sections are ellipses.

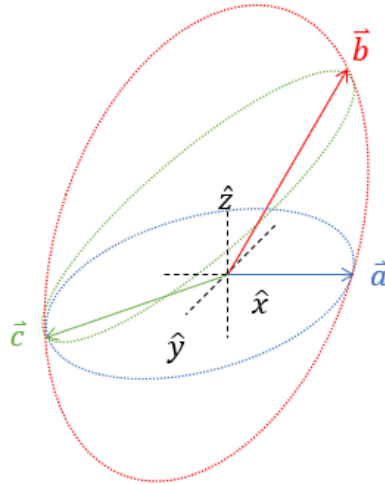
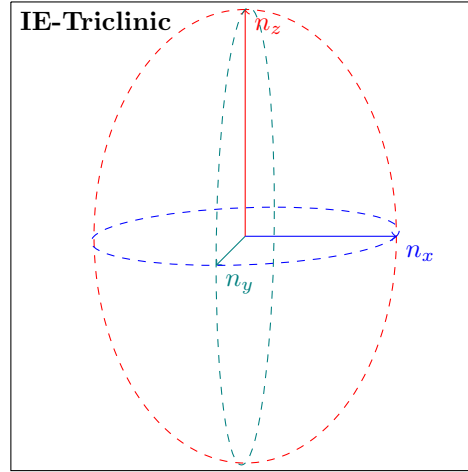


Figure 11: An oriented triaxial ellipsoid of the triclinic crystal.

For instance, in the red ellipse cross-section, $\vec{b}$ and $\vec{a}$ acted as the non-perpendicular semi-major axis and the semi-minor axis, respectively.

Now, take a different trail, and focus on principal axes. From the discussed equation, it is clear that we will get a triaxial ‘index’ ellipsoid for triclinic, which has semi-axes as the principal axes (mutually orthogonal but not equal), thus all those cross-sections would be again ellipses only. Let’s draw the required figure. For instance, in the red ellipse cross-section, n_z and n_x are the semi-major axis and the semi-minor axis, respectively. The figure represents the index ellipsoid(IE) of a triclinic crystal.



So far so good! But, I also want to see how the triclinic edge vectors $(\vec{a}, \vec{b}, \vec{c})$, which are *not mutually orthogonal*, but cross-section are circular or elliptical (depends upon, considering the one, out of first two drawn figures) and the principal axes (n_x, n_y, n_z) of the index ellipsoid of triclinic, which are *mutually orthogonal*, but are elliptical cross-section look together, when superimposed, because I do not foresee a one-to-one connection between these two parameters (yet), and thus can’t say whether both will align or overlap in the case of triclinic or not.

We already found the direction of the principal axes, but those are not known yet with reference to the edge vector, vice versa. Also, we don’t know the optical axis (axes) with reference to two sets of axes.

At first glance, it appeared to me that there should be no optical axis (OA) in triclinic, as it is very hard to find the axis of symmetry. I was regarding OA as the axis of symmetry, because from the earlier discussion on the optical axis, it appeared that OA is the axis of symmetry of the crystal system. Upon verification of my supposition, I realised the above supposition is incorrect.

35. So, what is the Optical axis of a triclinic crystal? If it exists, how to recognise it and also how to know where the edge vector lies with respect to the principal axis?
36. Let’s investigate. Firstly, name the triclinic edge vectors $(\vec{a}, \vec{b}, \vec{c})$ as **Crystallographic axes**, and (n_x, n_y, n_z) is already named as optical principal axes of the index ellipsoid.

Now, let me be a bit honest here, if we do a self-inquiry, I do not see any valid reason why nature should not be choosing beauty. By her preference for beauty, I mean the following: We

have 3D shapes, and we need to choose any two. The first figure is just three spheres embedded, each with a different radius and would not have having smooth boundary. Therefore, we are left with the last two figures (both having elliptical cross-sections), which are two triaxial ellipsoids; the first is associated with the Crystallographic axes and the other with the principal axes (I assume (intuitively) both would have some size and shape and would overlap, if carefully superimposed (see Fig.12), orienting anyone at a unique angle). To understand this more deeply, I see no constraints in the procedure of superimposing both. Once we achieve superimposition of them, we can visually see the required angle, say δ . At this stage, I would like to share an expression which I saw in the book [Sommerfeld, Optics],

$$\tan \delta_s = \frac{u_1}{u_3} \sqrt{\frac{u_2^2 - u_3^2}{u_1^2 - u_3^2}}$$

Where $u_1 > u_2 > u_3$ and u_i is the principal light velocity (along the principal axis i). I am taking a leap of faith saying $\delta \equiv \delta_s$; I may be utterly wrong, because the concepts around this stuff are yet to sink in.

Finally, the figure is,

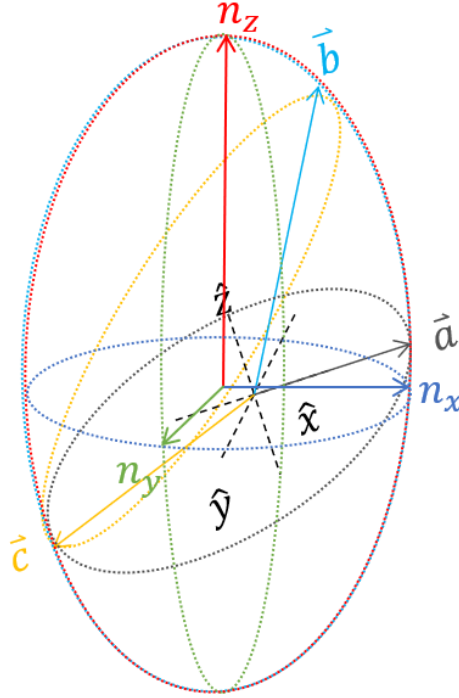


Figure 12: Superimposed

So far, we clearly see that all the highlighted cross-sections are ellipses and not circles.

From literature or earlier discussion on a uniaxial crystal, one can correctly realise that the optical axis is the line perpendicular to the circular section, instead of considering it as the axis of symmetry.

Once we agree on this definition of the optical axis, we need to search for a circular section. Just like we did in the tetragonal crystal discussion, there the basal plane was that circular section, and the line perpendicular ($\hat{z}$) was the same as the optical axis. Here, in triclinic, we don't see a circular basal plane, but can't that also mean no circular plane at all? I don't think so. To search for a circular section, we present a simple argument: In triclinic, $n_z > n_y > n_x$, and these are being treated as semi-axes, where n_z offers the highest RI, so the lowest phase velocity, so we call it the slow axis, and n_x offers the lowest RI, the fastest phase velocity, so we call it the fast axis, and n_y lies intermediate, between fast and slow axes. Let's pick a point on the zx plane, more specifically, on the red ellipse cross-section, having magnitude n_y , then we see there will be two such dimetrically opposite points (first and third quadrant). Now, tracing a plane by joining those two identified points along with another two intersection points (of teal cross-section and blue cross-section), we can finally get a circle (yellow). In the same manner, we can also get another circle (magenta), this time by picking mirror points (say, zy plane, a mirror plane). Altogether, we see two circular cross-sections; thus, the line perpendicular should be called the optical axis. Thus, in a triclinic crystal we got two optical axes (OA_1 and OA_2), thus it is a **biaxial crystal**. See fig.13.

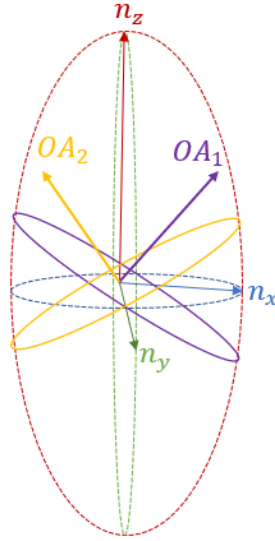


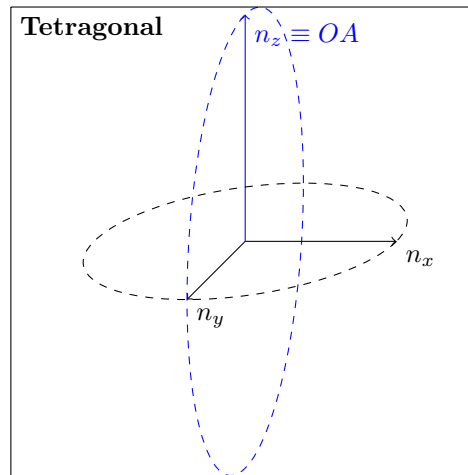
Figure 13: Biaxial Crystal

Having this also highlights that our previous consideration on taking ellipsoid based on crystallographic axes was irrelevant.

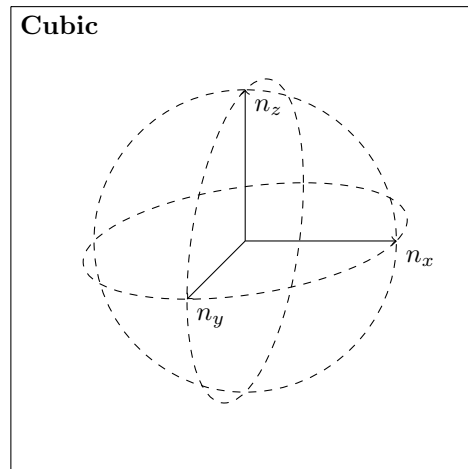
37. *What about other crystal systems?*

38. Let's talk about tetragonal and cubic.

For anisotropic (tetragonal), $n_x = n_y \neq n_z$, and all are mutually perpendicular, but the two basal planes are merged into circles, supporting o-rays, because their RI have n_o . On the other hand, the direction of OA forms an elliptical plane whose semi-minor axis is $n_x = n_y$ and semi-major axis is n_z , which supports e-ray, and we may say overall, the form is a "good-looking" ellipsoid. The crystal can be said to have one optical axis; therefore, it is a uniaxial crystal, and the OA can be made aligned to n_z . Overall, it exhibits cylindrical symmetry around the optical axis. This was a reason that wrongly motivated me to take OA as the axis of symmetry always.



For isotropic (cubic), $n_x = n_y = n_z$, and all are mutually perpendicular circular planes and exhibit spherical symmetry, and can be said to have infinite optical axes or every direction holds the same optical properties, in standard, we regard isotropic as having no OA. And each supports o-ray, hence no e-rays, thus only the common refraction phenomenon is seen.



39. So far, we have fixed our perspective on crystal, and noticed how refractive index varies in a crystal with direction, as indicated by the index ellipsoids of certain crystal structures. What if we see the same thing from the wave propagation point of view?
40. Yes. There is the standard way to depict the index ellipsoid, also known as the optical indicatrix, where we consider the propagating wave surface and monitor how it behaves in a different direction.

See fig.14. Suppose you have kept the light source at the origin, then since ordinary waves travel uniformly in every direction thus its wavefront will be spherical, and extraordinary waves show sensitive behaviour with direction, thus will trace an ellipsoid. However, only along the optic axis, both e-ray and o-ray experience the same RI, and thus will be in touch with one another.

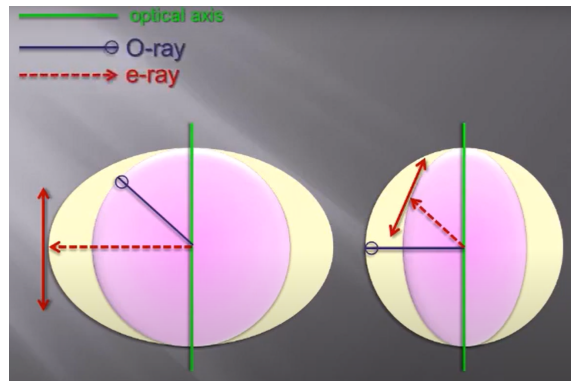


Figure 14: Wave propagation (Youtube)

In the figure, also notice that the polarisation state of the o-ray is orthogonal to the e-ray, as it is vertically polarised. And since $\vec{P}$ is always normal to the direction of wave propagation $\vec{k}$, any arrow from the centre is just $\vec{k}$, showing how it gets affected in every direction, if provided with an o-ray and an e-ray light source.

Now, in the left subfigure, we notice the e-ray travels faster than the o-ray, so it means the e-ray is experiencing lower RI compared to the o-ray, thus $n_e < n_o$, thus a negative crystal would be showing negative birefringence. Also, we regard the e-ray travelling on the fast axis. Things are opposite for right birefringences, in fact, it shows positive birefringence type.

Finally, it is interesting to note that when we shifted our perspective from crystal to wave, we somehow made things into a 2D abstract plot from 3D real geometry.

41. What do the ellipse and circle represent?
42. They are cross-sections of the index ellipsoid (optical indicatrix).
43. What do they show physically?
44. They represent how the refractive index depends on the direction of the wavevector $\vec{k}$.

For a given propagation direction (take $\vec{k}$ from the centre of the figure), the refractive index experienced by the wave is found from where $\vec{k}$ intersects the indicatrix (or boundary), which also gives us a unique angle with respect to the optical axis.

In a uniaxial crystal, we see that a circular outline (basal plane) corresponds to the ordinary index n_o , which is the same for all directions perpendicular to the optical axis.

The ellipse (cross-section including the optical axis) shows the extraordinary index $n_e(\theta)$, which varies smoothly with angle θ relative to the optical axis.

Thus, the index ellipsoid encodes how n_o or n_e vary with $\vec{k}$ direction.

45. *What one can achieve from the discussions made so far, say on light in free space, light in medium and getting acquainted with index ellipsoid or say double refraction or dispersion?*
46. In the real world, light, aka electric polarisation, is beautiful. It becomes more glorious when it goes into medium, it can produce or trigger various wave mixing processes, which may be beneficial from an application point of view and can be linear or non-linear, based on how the crystal responds. One such process is spontaneous parametric down-conversion (SPDC), which is scheme to get entangled photons, but once output are generated, they suffer from phase mismatch issue, due to dispersion, so to counter it, we may invoke another optical phenomenon, birefringence, which can heal the issue, otherwise one would have needed infinitely long crystal (in a hope to seek coherence), so better using the birefringent crystal, we can determine an phase matching angle through the analysis of index ellipsoid, which tell us at what angle incident pump (input) should be needed to enter the non-linear anisotropic media with respect to optical axis, so that generated photons are phase matched in SPDC. Achieving a phase-matched output is important because it indicates high-efficiency generation, which is required in the real world. Later, we consider those valuable quantum resources (phase-matched entangled photons) for quantum computation or apply them in quantum protocols, like quantum teleportation. That's one side of the coin. Additionally, one can build optical devices via a crystal, which behave as if they are a unitary operator or say quantum logic gates or detectors or projectors for photon states, when encoded in polarisation. Thus, it paves the way towards quantum computing and communication.
47. *So if one says that a non-linear process affected by phase mismatch $\Delta\vec{k} \neq 0$, due to dispersion $n(\omega)$, can be cured by birefringent phase matching, which implies the fact $n(\theta, \hat{p})$, is that correct?*
48. Yes. More correctly, for e-ray $n(\theta) \equiv n_e$ and represents one of the polarised light, and ordinary ray is just n_o and independent of any such consideration of θ . Then, birefringent phase matching, where to solve the dispersion problem we use birefringent crystal as a trick, where we look for condition $n_e(\theta, \omega_1) = n_o(\omega_2)$ which in disguise represents or implise $\Delta\vec{k} = 0$, thus by analysis of index ellipsoid deliver us a unique θ , which can be seen by noticing touch between circle and an ellipse in the index ellipsoid. It may be possible that there may be a mutual or distinct combination of touch between a circle or an ellipse.

Via AI,

One can get a slightly polished phrasing of the above point, which would be:

Yes. More correctly, the e-ray has an angle-dependent refractive index $n_e(\theta)$ while the o-ray is simply n_o , angle-independent. Birefringent phase matching exploits this fact: by choosing the

right propagation angle θ , we can satisfy conditions like $n_e(\theta, \omega_1) = n_o(\omega_2)$. This condition implies $\Delta \vec{k} = 0$, i.e. perfect phase matching. In the index ellipsoid picture, this appears as a tangency between the circle (ordinary surface) and the ellipse (extraordinary surface). Depending on the nonlinear process, there may be distinct or multiple such tangency points, corresponding to different types of phase matching (Type-I or Type-II).

49. It seems like we should go into describing a nonlinear optical process, and why phase matching is needed and how it is achieved, what if it is not achieved, and how it helps us to get the pattern on screen once phase-matched?

50. See next section.

0.3 Process, Effect and Generation

1. How is the Maxwell wave equation in media
2. We know from Ampere-Maxwell's law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\oint_{\partial \Sigma} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \left(\int_{\Sigma} \mathbf{J} \cdot d\mathbf{S} + \epsilon_0 \frac{\partial}{\partial t} \int_{\Sigma} \mathbf{E} \cdot d\mathbf{S} \right)$$

We know from Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\oint_{\partial \Sigma} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{\partial}{\partial t} \int_{\Sigma} \mathbf{B} \cdot d\mathbf{S}$$

We know from Stokes's theorem

$$\oint_{\partial \Sigma} \mathbf{A} \cdot d\boldsymbol{\ell} = \int_{\Sigma} (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

We know from Gauss's theorem

$$\oint_{\partial \Omega} \mathbf{A} \cdot d\mathbf{S} = \int_{\Omega} (\nabla \cdot \mathbf{A}) dV$$

Now, taking the partial derivative of the integral form of Ampere-Maxwell's law,

$$\oint_{\partial \Sigma} \frac{\partial \mathbf{B}}{\partial t} \cdot d\boldsymbol{\ell} = \left(\mu_0 \int_{\Sigma} \frac{\partial \mathbf{J}}{\partial t} \cdot d\mathbf{S} + \mu_0 \epsilon_0 \int_{\Sigma} \frac{\partial^2 \mathbf{E}}{\partial t^2} \cdot d\mathbf{S} \right)$$

Replacing the $\frac{\partial \mathbf{B}}{\partial t}$ using the differential form of Faraday's law,

$$-\oint_{\partial \Sigma} (\nabla \times \mathbf{E}) \cdot d\boldsymbol{\ell} = \int_{\Sigma} \left(\mu_0 \frac{\partial \mathbf{J}}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \right) \cdot d\mathbf{S}$$

Applying the formula of Stokes' theorem on LHS,

$$-\oint_{\Sigma} \nabla \times (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = \int_{\Sigma} \left(\mu_0 \frac{\partial \mathbf{J}}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \right) \cdot d\mathbf{S}$$

We get,

$$-\nabla \times (\nabla \times \mathbf{E}) = \left(\mu_0 \frac{\partial \mathbf{J}}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \right)$$

Using $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, we get,

$$-\nabla(\nabla \cdot \mathbf{E}) + \nabla^2 \mathbf{E} = \left(\mu_0 \frac{\partial \mathbf{J}}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \right)$$

Substituting, $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$, $\square = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$ and On rearranging,

$$\square \mathbf{E} = - \left(\frac{\nabla \rho}{\epsilon_0} + \mu_0 \frac{\partial \mathbf{J}}{\partial t} \right)$$

We have obtained inhomogeneous wave equations for the electric field. If $\rho = 0$ and $\mathbf{J} = 0$ then, homogeneous wave equations for the electric field be

$$\square \mathbf{E} = 0$$

and if not, then we retain,

$$\square \mathbf{E} = - \left(\frac{\nabla \rho}{\epsilon_0} + \mu_0 \frac{\partial \mathbf{J}}{\partial t} \right)$$

3. What if we replace the RHS term by Polarisation terms using relations $J_b = \frac{\partial \mathbf{P}}{\partial t}$ and $-\nabla \cdot \mathbf{P} = \rho_b$?
4. It is an important step towards getting the Maxwell equation inside the media, as we bring the polarisation $\vec{P}$, which is closely linked with the medium's property. However, we will consider the crystal not having a bound charge, $\rho_b = 0$, thus $\frac{\nabla \rho}{\epsilon_0}$ will vanish.

Upon such substitution, we see,

$$\square \mathbf{E} = \mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2}$$

5. We know that in component form,

$$P_i = \epsilon_0 \chi_{ij}^{(1)} E^j + \epsilon_0 \chi_{ijk}^{(2)} E^j E^k + \epsilon_0 \chi_{ijkl}^{(2)} E^j E^k E^l + \dots$$

what if we consider the non-linear part only

6. We get a non-linear wave equation. Since, if separating the linear and non-linear part, $\vec{P} = \vec{P}_L + \vec{P}_{NL}$, and substituting this we can have the Maxwell equation for non-linear media, and further, if the crystal is non-magnetic $\mu \sim \mu_0$ but exhibits high order electric susceptibility

$$\epsilon = \epsilon_0 + \epsilon_0\chi^{(1)} + \epsilon_0\chi^{(2)} + \epsilon_0\chi^{(3)} + \dots$$

that is, electric permittivity is tensorial, then,

$$\square \mathbf{E} \equiv \nabla^2 \mathbf{E} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \text{ So,}$$

Maxwell's wave equation for non-linear media,

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}_{NL}}{\partial t^2}$$

It is noticeable that $\mu_0 \frac{\partial^2 \mathbf{P}_{NL}}{\partial t^2}$ is acting as a source term, thus introducing a non-linear effect, and $\vec{E}$ is the incoming light field, which triggers certain effects inside the crystal.

7. *What is the functional form of $\vec{E}$?*
8. As the one-dimensional solution of the above equation, we see the functional form of $E^{(\omega)}$ (no need for a vector sign), meaning that the light has angular frequency ω . Also, consider the solution along the z direction with complex magnitude $E_1 = \tilde{E}_1 e^{i\phi_1}$ and wavevector k_1 , then, then scalar solution be,

$$E^{(\omega)} = \frac{1}{2} \left[E_1 e^{i(k_1 z - \omega t)} + c.c. \right]$$

9. *How to solve the Maxwell equation in non-linear media, say describing the case of Second harmonic generation?*
10. To first understand the Second harmonic generation (SHG): Let me introduce a general process called sum frequency generation (SFG), which is one of the **three wave mixing** processes that can occur in nonlinear media. Where if we input two lights of say ω_1 and ω_2 , we will get an output wave as $\omega_3 = \omega_1 + \omega_2$, and if $\omega_1 = \omega_2 = \omega$, we come closer to understanding SHG, where this special case $\omega \rightarrow 2\omega$ or simply $\omega_3 = 2\omega$.

This motivates us to have two functional forms of driving fields to describe the SHG.

$$E^{(\omega)} = \frac{1}{2} \left[E_1 e^{i(k_1 z - \omega t)} + c.c. \right]$$

and

$$E^{(2\omega)} = \frac{1}{2} \left[E_2 e^{i(k_2 z - 2\omega t)} + c.c. \right]$$

and to solve the nonlinear Maxwell equation, we also need a functional form of $P_{NL}^{(\omega)}$ and $P_{NL}^{(2\omega)}$ along with $E^{(\omega)}$ and $E^{(2\omega)}$. And these four will be substituted into the following decoupled equations:

$$\nabla^2 \mathbf{E}^{(\omega)} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}^{(\omega)}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}_{NL}^{(\omega)}}{\partial t^2}$$

$$\nabla^2 \mathbf{E}^{(2\omega)} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}^{(2\omega)}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}_{NL}^{(2\omega)}}{\partial t^2}$$

Now to get functional form of P_{NL} , we recall its expansion upto 2nd order that is, $P_{NL} = \epsilon_0 \chi^{(2)} E^2$ and if we substitute the form of $E = E^{(\omega)} + E^{(2\omega)}$ via simple algebra and $\chi^{(2)} = 2d$ (d is a matrix). we see

$$P_{NL}^{(\omega)} = \frac{\epsilon_0 d}{2} \left[2E_1^* E_2 e^{i((k_2 - k_1)z - \omega t)} + c.c. \right]$$

$$P_{NL}^{(2\omega)} = \frac{\epsilon_0 d}{2} \left[E_1^2 e^{i2((k_2 - k_1)z - \omega t)} + c.c. \right]$$

Since we have all four functional forms needed to solve two decoupled nonlinear Maxwell wave equations. One gets,

$$\left[2ik_2 \frac{\partial E_2}{\partial z} - k_2^2 E_2 - \mu_0 \epsilon (-4\omega^2) E_2 \right] e^{ik_2 z} = \mu_0 \epsilon_0 d E_1^2 (-4\omega^2) e^{2ik_1 z}$$

I have not detailed out how I reached this equation: Let me give you a brief. We are interested in the second harmonic wave 2ω , so we will only solve Maxwell's equations concerning 2ω and ignore the fundamental wave's Maxwell equations. The LHS of the equation has two terms, in mathematical manipulation, the first term $\nabla^2 E^{(2\omega)}$, we will apply "Slowly varying approximation" $|\frac{\partial^2 E_2}{\partial z^2}| \ll |\frac{\partial E_2}{\partial z}|$, physically it means the process is weak or envelope (2nd order partial derivative) varying slowly compared to first order partial derivative (relates to wavelength or period) in space and time. And then we move to solve the second term in LHS and the RHS via inserting the required functional forms and letting mathematical operators act, lastly putting all together, we have the above result.

Now, we can further compactify the result by bringing the dispersion relation $k_2^2 = (n_2 \frac{2\omega}{c})^2$, which cancels the 2nd and 3rd term in LHS. Thus, we are left with the following result, provided we club all exponentials together as,

$$\frac{\partial E_2}{\partial z} = \frac{i\omega d}{n_2 c} E_1^2 e^{-i\Delta k z}$$

This is the solution of the non-linear Maxwell equation for the second harmonic.

We call $e^{-i\Delta k z} = \Phi(k)$ as **phase-matching function**. Where $\Delta k = k_2 - 2k_1$.

11. *Why do we call three-wave mixing?*
12. It is because the resultant or third wave is generated due to the mixing of three frequencies $\omega_3 = \omega_1 + \omega_2$ or $\omega_3 = \omega_1 - \omega_2$, or just because a nonzero second-order electric susceptibility is involved, $\chi^{(2)} E(\omega) E(\omega)$ term.
13. *Is there any use of $\Phi(k)$, will it affect anything if not get vanished? And it seems that during the SHG $\Delta k \neq 0$ by itself, what does that signify?*

14. On observing the following expression,

$$[\dots] e^{ik_2 z} = (\dots) e^{2ik_1 z}$$

We notice that the LHS and the RHS waves are not moving in phase. So it seems LHS has wavelength $2\pi/k_2$ and RHS has π/k_1 , so the oscillating field (red) and varying polarisation (green) are not in phase. This continuous change in the phase relationship between the field and polarisation would result in energy to $\langle \vec{E} \cdot \frac{d\vec{P}}{dt} \rangle$ from polarisation to the field and back to polarisation. And, this suggests a lack of coherence.

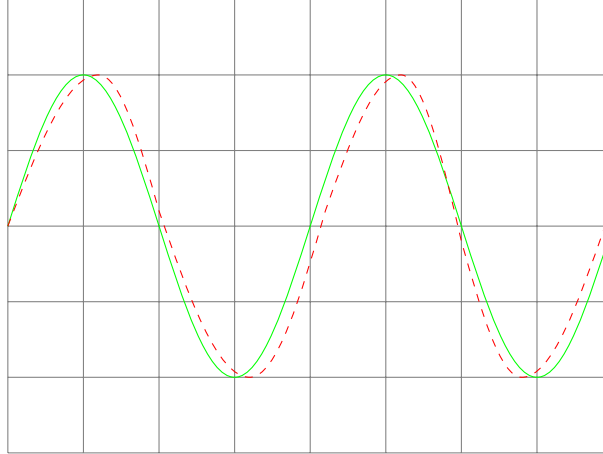
Basically, coherence length is defined as

$$L_c = \frac{\pi}{\Delta k}$$

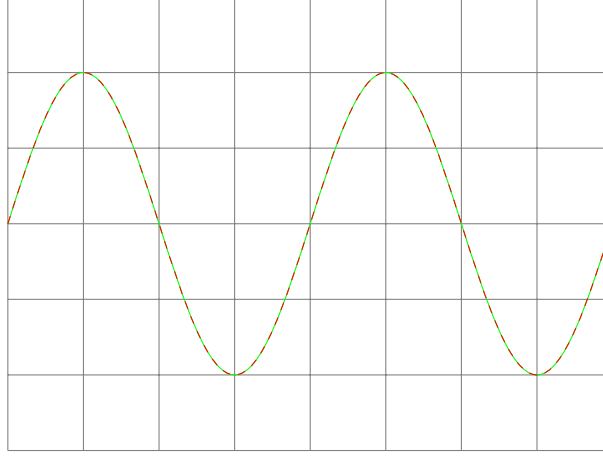
Having coherence suggests that when two waves move with the same phase velocity ($v_p(\omega_i) = \omega_i/k_i$), then there will be maximum energy transfer from polarisation to field, and this can only happen if $\Delta k = 0$, phase matched. So, one should intuitively think of phase matching as getting the same phase velocity of travelling waves.

Now, if one aims to get $\Delta k \rightarrow 0$, just by its own, then it basically wants $L_c \rightarrow \infty$, which indicates that we need an infinitely long crystal to get a phased match output, then that is something not practical to have. From the above expression, upon clubbing the exponentials, we define $\Phi(k)$, and having $\Phi \rightarrow 1$ is a crucial demand, which will ensure the maximum efficiency of wave mixing processes (such as SHG, SFG, and Difference frequency generation (DFG)).

Pictorially, we got something like this,



It signifies that the output is phased-mismatched, and the sole reason may be the phenomenon of normal dispersion. And we aim to get the following picture, and that without having an infinitely long crystal.



15. How is maximum efficiency related to phase-matching?

16. Provided we have

$$\frac{dE_2}{dz} = \frac{i\omega d}{n_2 c} E_1^2 e^{-i\Delta k z}$$

On solving the above equation with boundary values $E_2|_{z=0} = 0$; $E_2|_{z=z} = E_2(z)$, and in between defining compact trigometric function as $\text{sinc}(x) = \frac{e^{ix} - e^{-ix}}{2ix}$ We will see the result as,

$$E_2(z) = \frac{\omega d}{n_2 c} E_1^2 e^{-i\Delta k z/2} [zi \text{sinc}(\Delta k z/2)]$$

And interestingly $|E_2|^2$ relates to the intensity $I_2(z) = \frac{1}{2} \epsilon_0 n_2 c |E_2(z)|^2$ of output wave generation, which is further linked with power via $P_2(z) \propto I_2$. And finally, I can assure you these physical quantities, specifically the power, are measures of the efficiency of wave mixing generation, as the ratio of input and output power, i.e., $\eta = P_2/P_1$.

$$P_2(z) \propto P_1^2 \text{sinc}^2(\Delta k z/2)$$

Where P_1 corresponds to the power of the fundamental wave or first harmonic, and P_2 corresponds to the second harmonic.

So to have maximum efficiency, we need to mathematically and then repeat the same in the lab (practically), to look for the maximum value of $\text{sinc}^2(x)$.

So notice mathematically,

$$\lim_{\Delta k \rightarrow 0} \text{sinc}^2\left(\frac{\Delta k z}{2}\right) = z^2$$

So maximum power,

$$P_2(z)_{max} \propto z^2 P_1^2$$

So,

$$\eta_{max}(z) \propto z^2$$

And this was achieved only if $\Delta k \rightarrow 0$. See fig. 15.

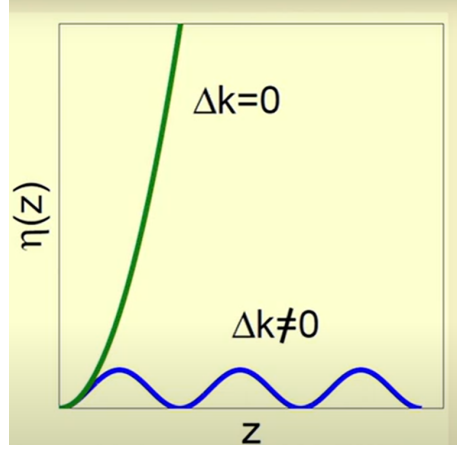


Figure 15: Efficiency plot (YouTube)

This was a mathematical task, but in the lab, how things will work out is a major problem.

17. How $\Delta k \rightarrow 0$ or aka 'getting the same phase velocities' in lab? And I don't see $\vec{k}$ as a kind of physical quantity one can control in a lab, and how to get convinced that geometry itself cannot resolve the issue of having $\Delta k = 0$?
18. Yes, it is right to say that $\vec{k}$ is directly not be made to work as desired. But let's see how $\Delta k = 0$ signifies,

$$\Delta k = 0 \implies k_2 - 2k_1 = 0 \implies k_2 = 2k_1$$

Now using the dispersion relation,

$$\implies 2\omega n_2 = 2\omega n_1 \implies n(2\omega) = n(\omega)$$

So it suggests that RI is the relevant parameter that can be considered to make phase matching. So, in expression-wise, what k does is basically done by n .

So it is better to call phase phase-matching condition

$$n(2\omega) - n(\omega) = 0$$

instead of $\Delta k = 0$, for the case of SHG.

So it has been the dispersion phenomenon which resulted in phase mismatch. This answers the middle part of the above question.

Now our target is answering the last part, before we answer the first part, which is mostly based on the description P.E. Powers gives in his book.

Suppose we consider a three-wave mixing process (instead of SHG), in an anisotropic crystal that exhibits normal dispersion. In such a crystal,

$$\omega_1 > \omega_2 \geq \omega_3$$

Then, normal dispersion says,

$$n_1 > n_2 \geq n_3$$

Now, $\Delta k \implies k_1 - k_2 - k_3$

then using the dispersion relation,

$$\frac{n_1 \omega_1}{c} - \frac{n_2 \omega_2}{c} - \frac{n_3 \omega_3}{c}$$

Since the wave mixing process must obey the energy conservation $\omega_1 = \omega_2 + \omega_3$.

Then,

$$\Delta k = \frac{1}{c} [n_1(\omega_2 + \omega_3) - n_2 \omega_2 - n_3 \omega_3] \implies \frac{1}{c} [\omega_2(n_1 - n_2) + \omega_3(n_1 - n_3)]$$

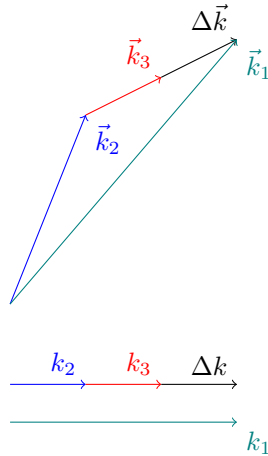
Now we have declared that $n_1 > n_2 \geq n_3$ therefore, always

$$\Delta k > 0$$

So, $k_1 > k_2 + k_3$

Thus, no closed geometry is possible as $\Delta k \neq 0$.

In figures,



Now, we are safe to answer the first part of the question: *How $\Delta k \rightarrow 0$ or aka ‘getting the same phase velocities’ in the lab, or ‘getting $n(2\omega) = n(\omega)$ ’ for the SHG case?*

We basically need to counter the normal dispersion $n(\omega)$ via birefringence $n(\theta, \hat{P})$ if we use a birefringent crystal. So carefully tuning the angle, or even other parameters such as temperature, may give us phase-matching in the lab.

In the lab, anisotropic crystals like BBO show double refraction, and this will make different $\vec{k}$ wave vectors to be either an o-ray or an e-ray. Thus making $n(2\omega) = n(\omega)$ possible .

19. *Explicitly show birefringent phase matching for SHG.*
20. We opt for a graphical method to show the birefringent phase matching explicitly. Since the SHG is basically,

$$\omega + \omega \rightarrow 2\omega$$

which we may equivalently call, $\omega_1 + \omega'_2 = \omega_3$ such that $\omega_1 = \omega'_2 \stackrel{\text{say}}{=} \omega_1$, also $\omega_3 \stackrel{\text{say}}{=} \omega_2$ and thus, $\omega_2 = 2\omega_1$. So, SHG is a degenerate wave mixing process. Since it is degenerate, two inputs are the same, which motivates us to employ an anisotropic (uniaxial) crystal to exploit birefringence for phase matching, which can be done by countering normal dispersion.

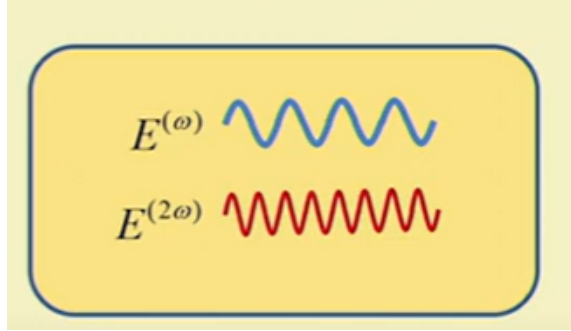


Figure 16: SHG inside crystal (Youtube)

In the picture, we have two waves $E^{(\omega)}$ and $E^{(2\omega)}$, and we ensure them to travel in phase. In fact, there are two such $E^{(\omega)}$, which acted as input to produce a $E^{(2\omega)}$.

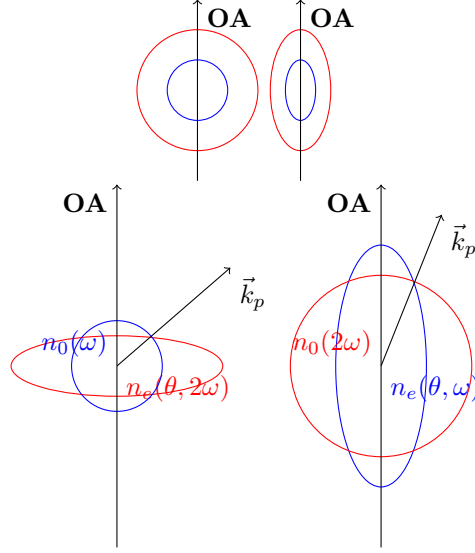
If we were not stuck to the SHG case, but any general three-wave mixing, then both inputs are allowed to have the same or different polarisation (o or e); but, only in SHG, they must be the same in polarisation, which restricts the number of logical combinations in the following list:

$$\begin{aligned} o(\omega) + o(\omega) &\rightarrow o(2\omega) \\ e(\omega) + e(\omega) &\rightarrow e(2\omega) \\ o(\omega) + o(\omega) &\rightarrow e(2\omega) \\ e(\omega) + e(\omega) &\rightarrow o(2\omega) \end{aligned}$$

We are well aware that o-ray corresponds to a circle and e-ray corresponds to an ellipse in optical index, and corresponding RI are $n_o(\omega_i)$ and $n_e(\theta, \omega_j)$. Such that,

$$n_o(\omega_i) \text{ (angle-independent), } \quad n_e(\theta, \omega_j) \text{ (angle-dependent), } \quad \frac{1}{n_e^2(\theta, \omega_j)} = \frac{\cos^2 \theta}{n_e^2(\omega_j)} + \frac{\sin^2 \theta}{n_o^2(\omega_j)}.$$

Therefore, the corresponding diagrams are for the above combination cases:



It is clearer that the first two cases would not make $n(\omega) = n(2\omega)$, as no intersection points exist. And the last two cases may cause things to happen at a certain angle θ_p , as phase matching corresponds to a ‘touching’ or intersection of two curves [On the optical indicatrix ordinary: circle, extraordinary: ellipse] and thus ensures $n(\omega) = n(2\omega)$; now the incoming wavevector $\vec{k}_p$ from origin passes the intersection point, it will make an phase matching with respect to OA, which ensure phase matched.

The third case is,

$$\begin{aligned} o + o \rightarrow e : \quad & 2k_1^o(\omega) = k_2^e(2\omega) \\ \Rightarrow 2 \frac{\omega n_o(\omega)}{c} = \frac{2\omega n_e(\theta_2, 2\omega)}{c} \Rightarrow & n_o(\omega) = n_e(\theta_2, 2\omega) \end{aligned}$$

Or, the fourth case

$$\begin{aligned} e + e \rightarrow o : \quad & 2k_1^e(\omega) = k_2^o(2\omega) \\ \Rightarrow 2 \frac{\omega n_e(\theta_1, \omega)}{c} = \frac{2\omega n_o(2\omega)}{c} \Rightarrow & n_e(\theta_1, \omega) = n_o(2\omega) \end{aligned}$$

The last two are examples of Type-I phase matching, as the output has an orthogonal polarisation state to the input (degenerated). Specifically, the third case corresponds to a negative uniaxial crystal, and the fourth or last case corresponds to a positive uniaxial crystal.

21. *Is there any way we can phase match the first two cases, even though they fail natural Birefringence phase matching.*
22. Yes, they *can* be engineered by quasi-phase matching, but not by natural birefringent PM in typical cases, as no θ exists.
23. *Since we refer last two cases a Type-I, does it indicate there are Type-II or more as well?*
24. Yes, if we had the wave mixing as

$$e + o \rightarrow e \quad e + o \rightarrow o$$

Then,

We call it Type-II birefringent phase matching (BPM), Type-II as SHG has mixed o-ray and e-ray at the degenerate frequency ω . We will not discuss it.

25. *How do we mathematically (numerically) determine the θ_p , for SHG, in type-I?*
26. We try to use Sellmeier equations. for the third case (negative uniaxial)

$$\begin{aligned} \Delta k &= [k_2^e(2\omega) - 2k_1^o(\omega)] \\ \Rightarrow \Delta k &= 2\pi \left[\frac{n_{2e}(\lambda_2, \theta)}{\lambda_2} - 2 \frac{n_{1o}(\lambda_1)}{\lambda_1} \right] \end{aligned}$$

for the fourth case (positive uniaxial)

$$\Delta k = 2\pi \left[\frac{n_{2o}(\lambda_2)}{\lambda_2} - 2 \frac{n_{1e}(\lambda_1, \theta)}{\lambda_1} \right]$$

Where we solve (numerically) for specific $\theta \in [0, \pi/2]$ i.e θ_p , which gives an appropriate propagation direction or makes $\Delta k = 0$. This can be done by using the energy conservation expression and also substituting the RI expression of e-ray in the Sellmeier equation.

Let's stick to a negative uniaxial crystal for the type-I BPM:

Then, the energy conservation expression:

$$\frac{1}{\lambda_2} = 2 \frac{1}{\lambda_1} \Rightarrow \lambda_1 = 2\lambda_2$$

then,

$$\Rightarrow \Delta k = \frac{4\pi}{\lambda_1} \left[n_{2e} \left(\theta, \frac{\lambda_1}{2} \right) - n_{1o}(\lambda_1) \right]$$

Also, substituting the RI expression of e-ray

$$n_{2e}(\theta, 2\omega_1) = \left[\frac{\cos^2 \theta}{n_e^2(2\omega_1)} + \frac{\sin^2 \theta}{n_o^2(2\omega_1)} \right]^{-1/2}$$

As $2\omega_1$ relates to $\frac{2}{\lambda_1}$, but there is no point of carrying 2, since it is n is just function of λ . So, RI expression of e-ray be,

$$\Rightarrow n_{2e}(\theta, \lambda_1) = \left[\frac{\cos^2 \theta}{n_e^2(\lambda_1)} + \frac{\sin^2 \theta}{n_o^2(\lambda_1)} \right]^{-1/2}$$

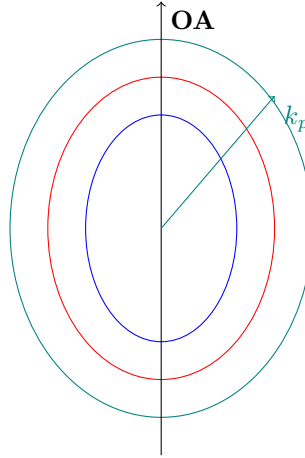
Then,

$$\Rightarrow \Delta k = \frac{4\pi}{\lambda_1} \left[\left[\frac{\cos^2 \theta}{n_e^2(\lambda_1)} + \frac{\sin^2 \theta}{n_o^2(\lambda_1)} \right]^{-1/2} - n_{1o}(\lambda_1) \right]$$

Now, solving the equation $\Delta k = 0$, would give us $\theta = \theta_p$. The solution(s) give the phase-matching angle(s) θ_p , then on the indicatrix, it indicates the circle-ellipse tangency point. Thus tell us, direction of $\vec{k}_p$, in and finally getting $\Delta \vec{k} = 0$.

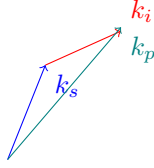
27. *Are there other parameters other than θ, λ ?*
28. Yes, in literature we see, not just only the choice of polarisation via incorporating λ and angle tuning via θ , but also Temperature tuning $n(T)$ exists. So together, $n(\theta, \lambda, T, \dots)$ is required. We will not talk beyond this.
29. *Can we solve this example, by finding two wavevectors angle, $\vec{k}_i$ and $\vec{k}_s$, such that*

$$\vec{k}_p - \vec{k}_s - \vec{k}_i = 0$$

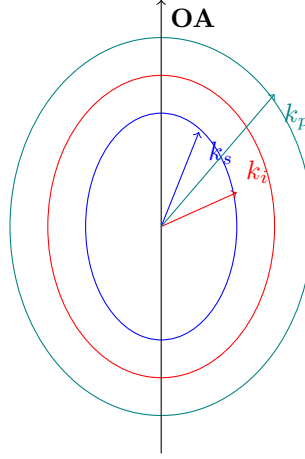


30. Let's try. We notice that these represent the e-ray, basically three e-rays. And no e-ray has zero path differences, since no touch between ellipses. It may express a process $e \rightarrow e + e$, where each ray is an e-ray. The gap depicts that the input and generated rays are experiencing different strengths of refractive index; at the same time, the input e-ray travels a larger distance, and the output travels a shorter distance. This can be explained by dispersion $n(\omega)$, which is further linked to the frequency or wavelength of the wave under consideration.

We aim to make a closed geometric figure, via respecting $\vec{k}_p = \vec{k}_i + \vec{k}_s$.



With such an objective, let's fit everything in the optical indicatrix.

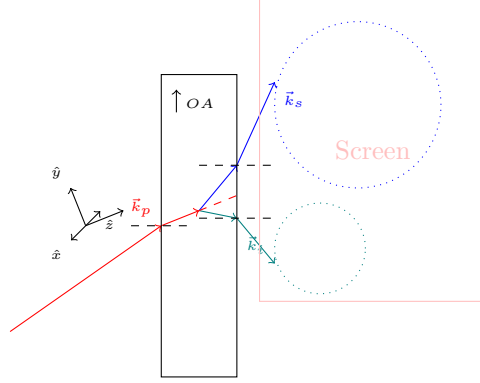


Since vector $\vec{k}_i$ is not able to touch the red ellipse, it shows that any rotation of k_i will not be able to make $\Delta\vec{k} = 0$

Therefore, this shows the **problem**, that it is not possible to phase-match $e \rightarrow e + e$ via a birefringence. However, it may be possible to phase-match through other schemes.

31. *How will after-phased matched outputs appear on the screen? Let's not restrict to the SHG process.*
32. As hinted, say, a pump photon k_p goes inside the anisotropic crystal, it responds nonlinearly, and thus, splitting of the frequency occurs via three-wave mixing and two sub-frequencies are generated.

We express it through a sketch: Where we are considering the phase-matched case of a process, $\omega_p \rightarrow \omega_s + \omega_i$. And generated frequencies do not need to be the same, but should obey energy conservation $\omega_p = \omega_s + \omega_i$ and wavevectors also different $\vec{k}_p - \vec{k}_s - \vec{k}_i = 0$. So let's call the angle between the normal and vector $\vec{k}_p$ as the angle of incidence θ_{inc} , the angle between the $\vec{k}_p$ and OA is the phase matching angle θ_p . Let the angle between the dashed red line and the blue line (inside) be θ_s and the angle between the dashed red line and the teal line (inside) be θ_i . Note that the dashed red line is along the $\hat{z}$, and we are seeing inside the front vertical cross-section of the crystal, which lies on the yz plane. The depth of the crystal lies on the xy plane. We put the scree parallel to the exit face of the crystal (xy plane). However, in the sketch, we showed it differently (yz plane) for a better view.



Each photon will show a circle on the screen, depending on whether the rotational symmetry about the azimuthal angle exists or not. It is satisfying to know that the process is independent of the azimuthal angle ϕ_i or ϕ_s . These angles may be connected via $\phi_i = \phi_s + \pi$ and lie on the xy plane. The independence of ϕ introduces the rotational symmetry, and all k_i and k_s will be uniformly distributed photons on the screen as a circle.

Through the following calculation, we show that these circles are just a projection of cones on the screen.

Vectors can be represented in spherical coordinates embedded in the Cartesian coordinate system,

$$\vec{k}_p = (0, 0, k_p)$$

$$\vec{k}_i = k_i(\sin \theta_i \cos(\phi_s + \pi), \sin \theta_i \sin(\phi_s + \pi), \cos \theta_i) = k_i(-\sin \theta_i \cos \phi_s, -\sin \theta_i \sin \phi_s, \cos \theta_i)$$

$$\vec{k}_s = k_s(\sin \theta_s \cos \phi_s, \sin \theta_s \sin \phi_s, \cos \theta_s)$$

Since, considered process is phase-matched, $\Delta \vec{k} = \vec{k}_p - \vec{k}_s - \vec{k}_i = 0$, therefore,

$$\Delta \vec{k} = (0 - k_s \sin \theta_s \cos \phi_s + k_i \sin \theta_i \cos \phi_s) - (0 - k_s \sin \theta_s \sin \phi_s + k_i \sin \theta_i \sin \phi_s) - (k_p - k_s \cos \theta_s - k_i \cos \theta_i) = 0$$

From z-axis components (magnitude-wise),

$$k_p^2 = k_s^2 \cos^2 \theta_s + k_i^2 \cos^2 \theta_i$$

This implies an equation of a circle.

and the other two components along the x-axis and the y-axis are independent of respective ϕ ,

$$k_s \sin \theta_s \cos \phi_s = k_i \sin \theta_i \cos \phi_s$$

Taking the square both sides,

$$k_s^2 \sin^2 \theta_s = k_i^2 \sin^2 \theta_i$$

using trigonometric identity $\sin^2 x = 1 - \cos^2 x$, then, we see,

$$k_s^2 - k_s^2 \cos^2 \theta_s = k_i^2 - k_i^2 \cos^2 \theta_i \implies k_s^3 - k_i^2 = k_s^2 \cos^2 \theta_s - k_i^2 \cos^2 \theta_i$$

Now, we have two equations, $[k_p^2 = k_s^2 \cos^2 \theta_s + k_i^2 \cos^2 \theta_i]$ and $[k_s^3 - k_i^2 = k_s^2 \cos^2 \theta_s - k_i^2 \cos^2 \theta_i]$
Solving the simultaneous equation,

we get,

$$\theta_s = \cos^{-1} \left(\frac{k_p^2 + k_s^2 - k_i^2}{2k_s^2} \right)$$

and also,

$$\theta_i = \cos^{-1} \left(\frac{k_p^2 - k_s^2 + k_i^2}{2k_i^2} \right)$$

These two are equations of a **cone** with respective half-cone angles $(\theta_s$ and $\theta_i)$ for corresponding photons $(\omega_s$ and $\omega_i)$.

Here at this stage, one can use dispersion relations $k = \frac{n\omega}{c}$, then,

we get,

$$\theta_s = \cos^{-1} \left(\frac{n_p^2 \omega_p^2 + n_s^2 \omega_s^2 - n_i^2 \omega_i^2}{2n_s^2 \omega_s^2} \right) \quad \theta_i = \cos^{-1} \left(\frac{n_p^2 \omega_p^2 - n_s^2 \omega_s^2 + n_i^2 \omega_i^2}{2n_i^2 \omega_i^2} \right)$$

And, to make things easy, if we consider the **degenerate** case $\omega_i = \omega_s = \omega$. And then, $\omega_p^2 = 4\omega^2$

we get, (let's do for signal photon γ_s),

$$\theta_s = \cos^{-1} \left(\frac{4n_p^2 \omega^2 + n_s^2 \omega^2 - n_i^2 \omega^2}{2n_s^2 \omega^2} \right) = \cos^{-1} \left(\frac{4n_p^2 + n_s^2 - n_i^2}{2n_s^2} \right)$$

This shows that θ_s just depends only on the refractive index that the pump, signal and idealer photons experience. This answers an unasked question: why does birefringent phase matching require involvement of RI via e-ray and o-ray?

33. *Can we make such an arrangement or phase matching that on screen we get two lines or even just two dots?*

34. Thinking....

Acknowledgement

I am thankful to my supervisor, Dr. Sandeep K. Goyal, for guidance, and my colleague, Dr. Satish Seth and Mr. Akhil Deswal. All of them have been encouraging me to visualise the physics, to ask questions in sequences and try answering them as far as possible. It brings clarity and makes the mind devoid of confusion. Reading books, watching lectures and interacting with generative AI may be a great way to collect information, but understanding is built only when we do directed thinking and express the same in a certain way. Thank you again.